

Software Requirements Specification

Campus Guide — One Platform for Campus Life, Career Guidance & Peer-Mentored Academic Help

"One campus-verified login, three connected experiences: stay informed, choose a path, get mentored."

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This document specifies the functional, non-functional, and data requirements of the Campus Guide platform: a campus-verified system uniting the Campus Info Hub, the Career Path Finder, and Mentor-Powered Help secured by bKash escrow.

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Chapter 1 — Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the complete software requirements of **Campus Guide**, a campus-verified web platform for United International University (UIU). The platform unifies three pillars of student life — official campus information, structured career direction, and paid, protected peer mentoring — into a single trusted system accessible through one university-verified login.

This document is intended for the project team, faculty supervisors, university administrators, student focus groups, and any future maintainers of the platform. It serves as the authoritative statement of what the system shall do (functional requirements), the qualities it shall exhibit (non-functional requirements), the data it shall manage, and the criteria against which it shall be accepted and tested.

Key Objectives

- Consolidate official notices, events, achievements, and emergency alerts into a single, always-current, searchable feed.
- Provide an interest- and preference-driven Career Path Finder that maps each student to a best-fit career track and a concrete roadmap.
- Enable students to book verified senior/peer mentors by course code, with payment held safely in bKash escrow until the session is confirmed.
- Guarantee trust through @uiu.ac.bd email verification, mentor vetting, ratings, transparent commission, and an admin-supervised dispute process.
- Operate as a free core utility (info hub and career finder) that sustains a lightweight, commission-based revenue model on mentor sessions.

1.2 Scope

Campus Guide is a web-based platform (frontend and backend both implemented in Next.js, backed by PostgreSQL and MongoDB) developed for the students, faculty, and administration of UIU. The scope of the release specified in this document covers the three pillars below.

Table 1.1 — In-scope product pillars

Pillar	Included Capability	Primary Users
Campus Info Hub	News, circulars, results, achievements, academic calendar, registration windows, exam dates, seminars, research and grant calls, role-based dashboards, emergency/safety alerts, and live traffic & weather near campus (Google Maps + OpenWeather).	All students, faculty, admins
Career Path Finder	Interest and job-preference assessment; matching engine that cross-references interests with target roles to surface a	Students (esp. underclassmen, undecided)

	best-fit career; personalized roadmap of courses, certifications, clubs, and mentor sessions.	
Mentor-Powered Help with bKash Escrow	Mentor discovery by exact course code; per-hour mentor pricing; session request/accept flow; bKash Merchant (Checkout/Tokenized) escrow payment; confirmation & two-way rating; admin-supervised dispute resolution with refund path; 8–12% session commission.	Students (seekers), mentors (seniors/peers), admins

Out of Scope

- Native mobile applications (the responsive web app and PWA shell cover mobile use).
- Automated AI-based tutoring content generation or proctoring.
- Direct bank/wallet integrations other than bKash in this release.
- Integration with the university's official student information system beyond email-domain verification.
- Multi-campus rollout beyond UIU (architecture remains multi-tenant-ready but only UIU is configured).

1.3 Definitions, Acronyms and Abbreviations

Table 1.2 — Definitions and abbreviations

Term	Expansion	Definition
SRS	Software Requirements Specification	This document; specifies requirements per ISO/IEC/IEEE 29148.
UIU	United International University	The institution the platform serves.
Mentor	—	A verified senior/peer student who offers paid academic help for specific course codes.
Seeker	—	A student who requests and pays for a mentor session.
Escrow	—	Funds captured by the platform via bKash and released to the mentor only after the seeker confirms delivery (or an admin resolves a dispute).
bKash	—	Leading Bangladeshi mobile financial service; Merchant Checkout / Tokenized checkout APIs process session payments.
Course Code	—	UIU course identifier (e.g., CSE-1109) used to match mentors to subjects.
MVP	Minimum Viable Product	The four-week pilot scope defined in the rollout plan (Section 14.3).
SSR	Server-Side Rendering	Next.js rendering strategy used for authenticated pages and SEO-sensitive feeds.

RLS	Row-Level Security	Database-level isolation of records by owning user.
RTM	Requirements Traceability Matrix	Mapping of requirements to use cases, design elements, and tests (Chapter 16).

1.4 References

- ISO/IEC/IEEE 29148:2018 — Systems and software engineering, Life cycle processes, Requirements engineering.
- IEEE 830-1998 (superseded practice, retained for structural convention).
- Campus Platform Proposal — "Campus Guide: One Platform for Campus Life, Career Guidance & Peer-Mentored Academic Help," August 2026 (primary functional source).
- "Comprehensive Career Guide with AI Model — SRS Document," UIU Department of CSE, December 2025 (structural and formatting reference).
- Next.js documentation — App Router and API Route Handlers (<https://nextjs.org/docs>).
- Prisma ORM documentation (<https://www.prisma.io/docs>) and Mongoose ODM documentation (<https://mongoosejs.com>).
- PostgreSQL 16 documentation (<https://www.postgresql.org/docs>); MongoDB 6+ manual (<https://www.mongodb.com/docs>).
- bKash Merchant APIs — Checkout and Tokenized Checkout documentation (developer.bka.sh).
- Google Maps Platform — Maps JavaScript API; OpenWeather API documentation.
- OWASP Application Security Verification Standard (ASVS) v4.0.3, Level 2; OWASP Top 10 (2021).
- W3C Web Content Accessibility Guidelines (WCAG) 2.1, Level AA.

1.5 Document Overview

The remainder of this document is organized as follows. Chapter 2 gives the overall description of the product, its users, environment, and constraints. Chapters 3–6 present the feasibility study, benchmark analysis, SWOT analysis, and stakeholder analysis that justify and shape the requirements. Chapter 7 enumerates the functional requirements and Chapter 8 models them as use cases. Chapter 9 specifies external interfaces and Chapter 10 the non-functional requirements. Chapter 11 documents the system models and data design; Chapter 12 consolidates constraints, assumptions, and dependencies; Chapter 13 analyzes risk. Chapters 14–16 cover the project plan, the test strategy and acceptance criteria, and the requirements traceability matrix. Chapter 17 contains appendices, including the database schema, UML models, wireframe inventory, API documentation, test cases, consultation notes, Gantt chart, and revision history.

Requirements are uniquely and stably identified: functional requirements as FR-<Area>-NN (e.g., FR-CI-03), non-functional requirements as NFR-<Quality>-NN (e.g., NFR-PE-02), use cases as UC-NN, and test cases as TC-NN. Within requirement tables, "shall" denotes a binding requirement; priorities follow MoSCoW (M = must, S = should, C = could).

Chapter 2 — Overall Description

2.1 Product Perspective

Campus Guide is a new, self-contained product — not a replacement module of an existing UIU system. Today, the functions it provides are scattered across informal channels: department notice boards, Facebook groups, word-of-mouth tutoring, and personal bKash transfers with no protection. Campus Guide replaces this fragmented landscape with one campus-verified platform in which the same trusted account powers information, career guidance, and paid mentor help.

The platform is a hosted Next.js application: the user interface and the API layer are both implemented in Next.js (App Router with server-side rendering and API route handlers). Two complementary databases are used: PostgreSQL (via Prisma ORM) for transactional data — identities, assessments, bookings, escrow transactions, ratings, disputes, and audit logs — and MongoDB Atlas (via Mongoose ODM) for document-oriented data — notices, events, mentor profiles, chat threads, and notification payloads. Redis provides session and rate-limit caching. External interfaces include the bKash Merchant API for escrow payments, Google Maps and OpenWeather for traffic and weather alerts, and university email (SMTP) for verification and notifications.

At a Glance	
Verified accounts	Only @uiu.ac.bd university email accounts may register
Core features in MVP	3 pillars: Info Hub, Career Path Finder, Mentor Help
Commission on sessions	8-12% per completed mentor session (below Fiverr's ~20%)
Build-to-pilot timeline	4-week MVP path inside the 16-week project schedule
Technology stack	Next.js (frontend + backend), PostgreSQL + MongoDB, Redis, bKash

The system context — the actors and external systems that exchange data with Campus Guide — is modeled in the context diagram (Figure 11.2) and the deployment view (Figure D.3).

2.2 Product Functions

At the top level, the product shall provide the following function classes:

Table 2.1 — Product function classes

Function Class	Summary of Behavior
Account & identity	Registration restricted to university email, email verification, password-based login with secure session management, role assignment (student, mentor, faculty, admin).
Campus Info Hub	

	Publication and consumption of notices, circulars, results, achievements, calendar items, seminars, research/grant calls; role-based dashboards; emergency alert broadcast; traffic and weather widgets.
Career Path Finder	Guided interest and job-preference assessment; matching against a curated career-path catalogue; personalized roadmap generation and tracking.
Mentor marketplace	Mentor onboarding and verification; course-code search; availability and per-hour pricing; session request, acceptance, delivery, confirmation, and rating.
Escrow payments	bKash Merchant payment capture into escrow at booking; release to the mentor (minus commission) on seeker confirmation; refund paths for decline, cancellation, and upheld disputes; frozen funds during disputes.
Trust & safety	Mentor verification by admins, ratings and completed-session counts, rate limiting, content moderation, and a complete admin audit log for every dispute resolution and refund.
Administration & reporting	Admin dashboards for dispute review, mentor approval, notice publication, commission configuration, and institution-level usage analytics.

2.3 User Classes and Characteristics

Table 2.2 — User classes

User Class	Characteristics	Technical Skill	Key Privileges
Students (general)	Every verified student; primarily consumes the Info Hub; mobile-first usage between classes.	Basic web/mobile literacy	Read feed, calendar, alerts; manage profile; take career assessment; book and rate mentors.
Career-seeking students	Underclassmen and undecided students choosing a track or specialization.	Basic	All student privileges; assessment and roadmap tracking emphasized.
Mentors (seniors/peers)	Students confident in specific courses; list availability, set hourly rates, earn a fee per session.	Basic to intermediate	Student privileges plus mentor profile, availability management, session acceptance/delivery, payout view.
Faculty / department staff	Faculty and administrative staff publishing official content.	Basic	Publish notices, events, achievements; (optionally) view aggregate, anonymized analytics for their department.
Administrators / moderators	Small trusted team operating the platform for the institution.	Intermediate	Verify mentors, approve events, publish emergency alerts, resolve disputes, configure commission, moderate content and accounts, view full analytics and audit logs.

2.4 Operating Environment

- **Client:** any modern evergreen browser (Chrome, Edge, Firefox, Safari, last two major versions) on desktop or mobile; the responsive UI targets ≥ 360 px viewports. No plug-ins required.
- **Server runtime:** Node.js 20+ on the Vercel platform (edge network + serverless functions), with TLS 1.3 enforced end-to-end.
- **Data stores:** PostgreSQL 16 (managed, e.g., Neon or Supabase), MongoDB Atlas 6+ (replica set), Redis 7 for cache.
- **External services:** bKash Merchant API (production and sandbox), Google Maps JavaScript API, OpenWeather API, and university SMTP/relay for email.
- **Network:** standard HTTPS over the public Internet; a student-grade connection (3G/4G/broadband) is assumed for acceptable performance.

2.5 Design and Implementation Constraints

- **C-1 Technology stack:** the frontend and backend shall both be implemented in Next.js (React, TypeScript); relational data shall use PostgreSQL and document data shall use MongoDB. This constraint is fixed by the team's existing skills and hosting plan.
- **C-2 Payment provider:** all session payments shall be processed exclusively through the bKash Merchant (Checkout/Tokenized) API; no direct card or other wallet support in this release.
- **C-3 Identity source:** registration is restricted to verified @uii.ac.bd addresses; the platform shall not operate an open public signup.
- **C-4 Hosting:** serverless hosting (Vercel) with managed databases; no institution-operated physical servers are available to the project.
- **C-5 Budget:** the platform must run within a student-project budget; paid tiers are limited to low-cost managed plans (see Section 14.5).
- **C-6 Schedule:** the team is scoped for the 16-week schedule in Chapter 14; the 4-week MVP pilot path bounds the first release.
- **C-7 Compliance:** security controls shall conform to OWASP ASVS Level 2; accessibility to WCAG 2.1 AA; and data handling to applicable Bangladeshi data-protection requirements.
- **C-8 Legal:** escrow-style fund handling shall be limited to capturing, holding, and releasing session payments against the merchant account — no wallet balances may be stored on the platform.

2.6 User Documentation

- **Student guide:** quick-start for feed, calendar, alerts, career assessment, and booking/rating a mentor (in-app help + web PDF).
- **Mentor guide:** onboarding, verification steps, availability and pricing setup, session delivery norms, and payout/fee explanation.
- **Faculty & admin manual:** notice publication, emergency alert broadcast, mentor approval, dispute resolution, and commission configuration procedures.

- **API documentation:** OpenAPI-style reference for all Next.js API routes (Appendix F summarizes the endpoints).
- **FAQ & dispute policy:** plain-language explanation of escrow, commission, refund, and dispute rules.

2.7 Assumptions and Dependencies

Table 2.3 — Key assumptions and dependencies

ID	Type	Statement
A-1	Assumption	UIU permits @uiu.ac.bd email-based verification and does not object to a student-run platform operating under the university's name with institutional branding approval.
A-2	Assumption	A pilot cohort of mentors and seekers volunteers in the first semester, seeded by the team.
A-3	Assumption	Faculty and staff will publish notices through the platform at least as often as through legacy channels during the pilot.
D-1	Dependency	bKash merchant account approval and sandbox credentials are granted before payment integration begins.
D-2	Dependency	Google Maps and OpenWeather API keys remain within free-tier or budgeted quotas.
D-3	Dependency	University SMTP relay (or equivalent transactional email service) is available for verification mail.
D-4	Dependency	Vercel, Neon/Supabase, and MongoDB Atlas managed services maintain their published uptime SLAs.

Chapter 3 — Feasibility Study

The feasibility of Campus Guide was assessed across five dimensions before requirements were fixed. Each dimension is summarized below with its verdict.

3.1 Technical Feasibility

Every element of the platform uses proven, well-documented technology already within the team's skill set. Next.js provides both the SSR frontend and the API layer in one codebase; Prisma and Mongoose give mature, type-safe access to PostgreSQL and MongoDB respectively. The bKash Merchant Checkout and Tokenized Checkout APIs are documented, sandboxed, and widely used in Bangladesh, so escrow capture and release are implementable without custom payment infrastructure. Google Maps and OpenWeather are simple REST integrations. The only moderately novel component — the career matching engine — is a transparent weighted scoring model over a curated catalogue, well within reach.

Table 3.1 — Technical feasibility assessment

Aspect	Assessment	Verdict
Frontend & backend (Next.js)	Single-language TypeScript stack, SSR, API routes; team already proficient	Feasible
Dual data stores (PostgreSQL + MongoDB)	Clear separation of transactional vs. document data; managed services remove ops burden	Feasible
bKash escrow orchestration	Sandbox available; capture/release/refund endpoints documented; IPN callbacks supported	Feasible (sandbox-first)
Career matching engine	Weighted interest/preference scoring over a curated catalogue; no ML infrastructure needed	Feasible
Real-time alerts (maps, weather)	Third-party REST APIs with generous free tiers	Feasible

3.2 Economic Feasibility

The platform is built by a student team, so development labor is course-funded. Recurring infrastructure cost is limited to low-tier managed services. Revenue arises from an 8-12% commission on completed mentor sessions and optional featured-mentor placement — the core Info Hub and Career Path Finder remain free to drive daily adoption. Even under conservative pilot volumes, commission income covers running costs.

Table 3.2 — Annual operating cost estimate (BDT)

Item	Est. Annual Cost	Notes
Hosting (Vercel Pro)	≈ 26,000	Team plan; Hobby tier is free during development

PostgreSQL (Neon/Supabase)	≈ 30,000	Low usage tier; free tier usable in pilot
MongoDB Atlas	≈ 0-70,000	Free M0 cluster in pilot; M2-M5 as volume grows
Domain + SSL	≈ 2,000	SSL itself is free (Let's Encrypt / platform-managed)
Email service	≈ 0-12,000	University SMTP or a low transactional tier
Maps / Weather APIs	≈ 0-10,000	Free tiers cover pilot usage
Total (pilot)	≈ 30,000-60,000 BDT/yr	Rises with volume, offset by commission

Break-even illustration: at an average session fee of 300 BDT and 10% commission, each completed session yields 30 BDT; roughly 200 sessions/month covers the upper pilot-year cost band — a modest target for a campus of UIU's size.

3.3 Operational Feasibility

- **Fit to real behavior:** students already live on their phones; the platform replaces channels they use today (Facebook groups, DMs, cash/bKash deals) rather than demanding new habits.
- **Low support load:** verification is automated via university email; moderation is scoped by rate limiting, audit logs, and a small admin panel usable by non-technical staff.
- **Champion model:** mentors earn visibly, giving seniors an incentive to promote the platform organically.
- **Training:** one short guide per user class (Section 2.6) is sufficient; no training infrastructure is required.

3.4 Schedule Feasibility

A 16-week schedule (Section 14.3, Figure I.1) leads from requirements sign-off to a pilot launch at UIU, with security and accessibility audits before UAT. Within it, the proposal's 4-week MVP path (foundation → career & booking → trust layer → polish & pilot) is preserved as the pilot-critical core, ensuring a usable product early even if later refinements slip.

3.5 Legal Feasibility

- **Payments:** escrow-style handling is limited to capture/hold/release against an approved bKash merchant account; no stored wallet balances, in line with bKash merchant terms.
- **Data protection:** PII is limited to what the university identity already implies; consent is captured at registration; retention and deletion rules are defined (Section 10.10).
- **Content:** notices are official university content published by authorized staff; user content is moderated under a published policy with an audit trail.

- **Intellectual property:** all third-party libraries are permissively licensed (MIT/Apache); no licensed assets are required.
- **Terms of service:** a plain-language ToS covering session conduct, refunds, and dispute arbitration is published at launch.

3.6 Feasibility Summary

Table 3.3 — Feasibility summary

Dimension	Verdict	Controlling Factor
Technical	Feasible	Proven stack; bKash sandbox-first integration
Economic	Feasible	≈ 30–60k BDT/yr pilot cost vs. commission upside
Operational	Feasible	Replaces existing habits; small admin workload
Schedule	Feasible	16-week plan with de-risked 4-week MVP core
Legal	Feasible	Merchant-account escrow model; published ToS and data policy

Overall verdict: the project is feasible on all five dimensions and proceeds to requirements specification.

Chapter 4 — Benchmark Analysis

4.1 Existing Systems

Students currently meet each pillar's need through a mix of informal and general-purpose systems. The benchmark set comprises: (1) **Facebook groups & department pages** for notices, (2) the **university website** for official circulars and calendar, (3) **LinkedIn / generic career advice sites** for career direction, and (4) **tutoring marketplaces** (Fiverr, Shikho, 10 Minute School) for paid academic help.

4.2 Feature Comparison

Table 4.1 — Feature comparison with existing systems

Capability	FB Groups	UIU Website	LinkedIn / Advice Sites	Tutoring Marketplaces	Campus Guide
Campus-verified accounts	✗	✗	✗	✗	✓ (@uiu.ac.bd)
Official notices in one feed	Partial	✓	✗	✗	✓
Emergency alert broadcast	Partial	✗	✗	✗	✓
Live traffic & weather near campus	✗	✗	✗	✗	✓
Structured interest-based career matching	✗	✗	Partial	✗	✓
Campus-specific career roadmap	✗	✗	✗	✗	✓
Mentor search by exact course code	✗	✗	✗	Partial	✓
Escrow-protected payment	✗	✗	✗	✓	✓ (bKash)
Ratings & completed-session record	✗	✗	Partial	✓	✓
Campus-level dispute resolution	✗	✗	✗	Partial	✓ (admin panel)
Free core utility	✓	✓	Partial	✗	✓

4.3 Strengths of Existing Systems

- Facebook groups have unmatched reach and immediacy among students and require no new signup.
- The university website is authoritative for official circulars and is publicly indexed.
- LinkedIn offers industry-scale networking; general advice sites provide broad career content.
- Tutoring marketplaces (Fiverr ~20% commission) demonstrate validated demand for paid help and provide mature payment and review mechanics.

4.4 Limitations

- Notices on social channels are scattered, unverified, and easy to miss; there is no single source of truth or emergency broadcast guarantee.
- The static website cannot push real-time alerts, has no personalization, and offers no engagement loop that brings students back daily.
- Generic career tools are not mapped to UIU's tracks, electives, clubs, or local job market, so advice stays abstract.
- Tutoring marketplaces are expensive for students, not course-code-accurate, outside campus trust circles, and lack an institution-backed dispute path.
- Nowhere can a student pay for peer help with funds held in escrow; informal bKash transfers carry full loss risk if a session falls through.

4.5 Proposed Improvements

- **One verified feed:** all official content plus emergency, traffic, and weather alerts in a single searchable, role-aware feed — replacing scattered group-scrolling.
- **Campus-calibrated matching:** the matcher cross-references stated interests and job preferences against a curated catalogue of UIU-aligned career paths, then turns the match into a concrete roadmap (courses, certifications, clubs, mentor sessions).
- **Course-code-accurate mentors:** seekers find seniors who actually took their course, with a visible rating and completed-session record before booking.
- **Protected economics:** bKash escrow with confirmation-triggered release, a 8–12% campus-competitive commission (vs. ~20% elsewhere), and an admin-supervised dispute path with refunds — never a silent loss for either side.

Chapter 5 — SWOT Analysis

5.1 Strengths

- Campus exclusivity: @uiu.ac.bd verification creates a trusted, spam-resistant community on day one.
- Three-pillar synergy: the free Info Hub and Career Finder build daily-login habit that feeds the monetized mentor marketplace.
- Escrow-protected payments differentiate the platform from every informal tutoring channel.
- Lean, proven stack (Next.js + PostgreSQL + MongoDB) owned end-to-end by a resident student team.
- Low break-even economics with commission income that scales with adoption.

5.2 Weaknesses

- Cold-start: the mentor marketplace needs a seeded cohort before search results look credible.
- Small team concentrates knowledge; bus-factor risk during exams and breaks.
- Dependence on third-party uptime (bKash, Vercel, Atlas, APIs) outside the team's control.
- Content freshness depends on faculty adopting the publishing flow.
- Single-campus scope limits early network effects; scaling to other campuses is unbudgeted.

5.3 Opportunities

- Expansion to other Bangladeshi universities once the UIU pilot proves the model (multi-tenant architecture).
- "Pro Mentor" verified-badge subscription and featured placement as additional revenue lines.
- Institutional licensing of analytics dashboards to departments and the career center.
- Alumni mentors and employer partnerships extending the mentor pool beyond current students.
- Bundled services around the roadmap (certification discounts, club events, workshop ticketing).

5.4 Threats

- Competing student projects or the university's own official app could duplicate the Info Hub.
- Regulatory or policy shifts in mobile financial services affecting escrow-style handling.

- Abuse vectors (fake disputes, spam listings, harassment) that could erode trust faster than moderation can respond.
- Data-protection obligations tightening around student PII.
- Graduation churn: each year's mentors and admins rotate out, requiring continual onboarding.

5.5 SWOT Matrix

Table 5.1 — SWOT matrix with strategic responses

Helpful (internal)	Harmful (internal)
Strengths Verified campus community; three-pillar synergy; escrow trust; lean proven stack; low break-even. Leverage: seed mentors from known seniors; publicize escrow story in launch communications.	Weaknesses Cold start; bus factor; third-party dependence; faculty adoption; single campus. Remedy: documentation and cross-training; sandbox fallbacks and circuit breakers for external APIs; launch partner in each department.
Opportunities Multi-campus expansion; Pro Mentor/featured revenue; institutional analytics; alumni mentors; roadmap bundles. Exploit: keep schemas tenant-scoped from day one; pilot analytics with the career center.	Threats Official-app duplication; MFS policy shifts; abuse vectors; PII obligations; graduation churn. Counter: position as the university's student-built partner; published moderation and data policy; admin handover kit each semester.

Chapter 6 — Stakeholder Analysis

6.1 Stakeholder Identification

Table 6.1 — Stakeholder identification

Stakeholder	Category	Stake in the Project
Students (general)	End user	Timely notices, calendar, alerts; free access to campus information.
Career-seeking students	End user	Structured, UIU-calibrated guidance for choosing a track.
Seekers (help-buying students)	End user	Protected payment and trustworthy mentors before mid/final exams.
Mentors (seniors/peers)	End user / beneficiary	Fair earnings, visibility, and protection from non-paying seekers.
Faculty / department staff	Content publisher	A reliable official channel that students actually read.
Administrators / moderators	Operator	Manageable moderation load, clear dispute tools, audit confidence.
University administration (UIU)	Institutional sponsor	Improved information flow, student outcomes analytics, brand goodwill.
Project team & supervisor	Developer / academic	Successful delivery of a course project and a live pilot.
bKash (merchant partner)	External provider	Transaction volume; merchant compliance.

6.2 Stakeholder Roles

- **Students/seekers:** consume content, take assessments, book and rate sessions; primary source of feedback during UAT.
- **Mentors:** supply the inventory of help; co-own the quality bar through ratings and completed counts.
- **Faculty/staff:** authoritative publishers; gatekeepers of official content accuracy.
- **Admins/moderators:** run trust & safety: verification, disputes, refunds, moderation, alerts.
- **University administration:** grants email-verification legitimacy and pilot permission; receives aggregate reporting.
- **Project team:** designs, builds, tests, and operates the platform through the pilot.

6.3 Stakeholder Interests

Table 6.2 — Interests and influence focus

Stakeholder	Primary Interest	Success Looks Like
Students	Never miss an important notice; get help safely before exams	Weekly active use; disputes rare and resolved fairly
Mentors	Predictable earnings and respectful treatment	Completed sessions paid out on time; visible track record
Faculty	Reach students without managing scattered groups	Notice views exceed Facebook-group equivalents
Admins	Low-effort, auditable moderation	Disputes resolved within 48h; clean audit log
UIU	Student success and institutional reputation	Pilot report: adoption, career-track choices, zero payment incidents
bKash	Compliant, low-friction transactions	No chargebacks or policy violations from the merchant account

6.4 Power-Interest Analysis

Table 6.3 — Power-interest grid and engagement strategy

Quadrant	Stakeholders	Strategy
High power · High interest (Manage closely)	University administration, project supervisor, admins	Weekly status, decision checkpoints at milestones, sign-off on ToS/policy
High power · Low interest (Keep satisfied)	bKash (merchant partner), department heads	Compliance reporting, quarterly summaries, escalation path
Low power · High interest (Keep informed)	Students, seekers, mentors, faculty publishers	Release notes, in-app announcements, feedback form, pilot cohort channel
Low power · Low interest (Monitor)	General public, other universities	Minimal effort; monitor demand signals for expansion

6.5 Communication Plan

Table 6.4 — Communication plan

Stakeholder	Channel	Frequency	Owner	Content
Supervisor	Meeting + progress doc	Weekly	Project lead	Sprint demo, risks, decisions needed
University admin	Email + milestone review	Bi-weekly / at milestones	Project lead	Adoption metrics, policy sign-offs
Admins/moderators	Admin panel notes + chat	Daily during pilot	Ops lead	Dispute queue, verification queue, alerts
Faculty publishers			Ops lead	

	Email + in-app notice	At onboarding, then as needed		Publishing guide, content calendar
Students & mentors	In-app announcements, email	Per release	Team	Feature notes, policy updates, outage notices
bKash	Developer portal / support	As needed	Project lead	Integration issues, compliance questions

Chapter 7 — Functional Requirements

This chapter enumerates the functional requirements of Campus Guide. Priorities use MoSCoW: **M** (must, required for pilot), **S** (should, planned but deferrable one iteration), **C** (could, enhancement). Every requirement is uniquely identified and traced in Chapter 16.

7.1 User Management

ID	Requirement	Priority
FR-UM-01	The system shall allow registration only with a valid @uiu.ac.bd email address; other addresses shall be rejected with a clear message.	M
FR-UM-02	The system shall send a verification link to the registered address and activate the account only after verification within 24 hours.	M
FR-UM-03	The system shall assign each user exactly one primary role (student, mentor, faculty, admin) and permit eligible students to additionally hold the mentor capability.	M
FR-UM-04	The system shall allow users to close their account; related personal data shall be anonymized per the retention policy (NFR-PR-02).	S
FR-UM-05	The system shall prevent duplicate accounts for the same university email.	M
FR-UM-06	The system shall display a user's role-appropriate dashboard immediately after login.	M

7.2 Authentication

ID	Requirement	Priority
FR-AU-01	The system shall authenticate users with email and password using a modern password-hashing function (bcrypt or Argon2) and shall never store plaintext credentials.	M
FR-AU-02	The system shall establish server-side sessions (or signed JWTs) with secure, HttpOnly cookies and a configurable idle timeout of 30 minutes.	M
FR-AU-03	The system shall enforce rate-limited login (max 5 failed attempts per 15 minutes per account/IP) and lockout with email-based unlock.	M
FR-AU-04	The system shall support password reset via time-limited, single-use emailed links.	M
FR-AU-05	The system shall protect all state-changing requests with CSRF tokens.	M
FR-AU-06	The system shall require re-authentication before sensitive actions (payout account changes, account closure).	S

7.3 Profile Management

ID	Requirement	Priority
FR-PM-01	Users shall be able to view and edit their profile: display name, department, intake/semester, avatar, and bio.	M

FR-PM-02	Students shall be able to maintain a list of interests (e.g., algorithms, UI design, networking, finance) and a preferred job role (e.g., Software Engineer, Data Analyst, Product Designer).	M
FR-PM-03	Mentors shall be able to create a mentor profile listing course codes they teach, per-hour fee, short description, weekly availability, and payout details.	M
FR-PM-04	Users shall be able to control notification preferences per category (notices, bookings, disputes, system).	S
FR-PM-05	Mentor profiles shall publicly display rating average, completed-session count, and verification badge; personal contact details shall remain private until a session exists.	M

7.4 Core Business Functions

7.4.1 Campus Info Hub

ID	Requirement	Priority
FR-CI-01	Authorized faculty/admin users shall be able to publish notices, circulars, results announcements, achievements, calendar items, seminars, and research/grant calls, each with type, priority, audience, publish time, and expiry.	M
FR-CI-02	Admins shall be able to publish an emergency alert that is pinned at the top of every dashboard, marked visually distinct, and pushed to all users within 60 seconds.	M
FR-CI-03	The feed shall be filterable by type, category, department, and priority, and searchable by keyword.	M
FR-CI-04	The system shall render role-based dashboards: students see the student feed; faculty see publishing tools and their department's content; admins see operational dashboards.	M
FR-CI-05	The system shall display live traffic conditions and weather near campus using Google Maps and OpenWeather data, refreshed at least every 15 minutes.	S
FR-CI-06	The system shall display the academic calendar with registration windows, exam dates, and event listings, and allow per-item reminders.	S
FR-CI-07	Expired notices shall be archived automatically and remain searchable.	S
FR-CI-08	Every notice shall record its author, timestamps, and edit history visible to admins.	S

7.4.2 Career Path Finder

ID	Requirement	Priority
FR-CP-01	Students shall be able to complete a guided assessment selecting enjoyed subjects/topics and a target job preference.	M
FR-CP-02	The matching engine shall cross-reference interests and job preferences against the curated career-path catalogue and produce a ranked list of best-fit careers with match scores.	M
FR-CP-03		M

	For a selected career, the system shall generate a personalized roadmap: recommended UIU courses and electives, certifications, clubs/communities, and suggested mentor sessions.	
FR-CP-04	Students shall be able to save a career path and track roadmap steps as not-started / in-progress / done.	M
FR-CP-05	The system shall allow re-assessment at any time, preserving previous results for comparison.	S
FR-CP-06	Roadmap recommendations shall be recalculated when a student updates interests or job preference.	S

7.4.3 Mentor-Powered Help with bKash Escrow

ID	Requirement	Priority
FR-MH-01	Mentor applicants shall submit a mentor profile that enters a verification queue; only admins can approve a mentor (verification checks course history and standing).	M
FR-MH-02	Seekers shall be able to search mentors by exact course code, optionally filtered by rating, fee range, and availability.	M
FR-MH-03	A seeker shall be able to request a session with a mentor for a course code, topic, preferred time, and duration; the request creates a MentorSession in state Requested.	M
FR-MH-04	On request, the system shall compute the total fee (hourly rate × duration) and initiate bKash payment; funds shall be captured into escrow, not transferred to the mentor.	M
FR-MH-05	A mentor shall accept or decline a request within 48 hours; on decline or timeout the escrowed amount shall be automatically refunded in full to the seeker.	M
FR-MH-06	After the session, the mentor shall mark the session delivered; the seeker shall confirm completion and submit a 1–5 star rating with optional comment.	M
FR-MH-07	On seeker confirmation, the system shall release the escrowed amount to the mentor minus the configured commission (8–12%), and record the payout transaction.	M
FR-MH-08	Either party shall be able to raise a dispute before confirmation; a dispute shall freeze the escrowed funds and open a case in the admin dispute panel.	M
FR-MH-09	Admins shall resolve disputes by ruling for the seeker (full/partial refund) or the mentor (release of funds); every resolution shall be written to the audit log with the deciding admin, rationale, and amounts.	M
FR-MH-10	The system shall support an auto-confirm window of 48 hours after delivery marking; if the seeker neither confirms nor disputes in that window, the session auto-confirms and funds release.	S

7.5 Search and Filtering

ID	Requirement	Priority
FR-SF-01		M

	The system shall provide a global search across notices, events, mentors, and career paths with keyword and category filtering.	
FR-SF-02	Mentor search results shall be rankable by relevance, rating, completed count, and fee.	M
FR-SF-03	Search shall return results within 2 seconds for the pilot dataset and shall paginate beyond 20 results.	M
FR-SF-04	Search queries shall be rate-limited per user to prevent scraping (NFR-SE-08).	S

7.6 Notifications

ID	Requirement	Priority
FR-NT-01	The system shall send in-app notifications for session state changes (requested, accepted, declined, funded, delivered, confirmed, disputed, refunded) to the affected parties.	M
FR-NT-02	The system shall send email notifications for account verification, password reset, booking lifecycle, dispute outcomes, and emergency alerts.	M
FR-NT-03	Emergency alerts shall additionally be pushed to all users via in-app banner on next load and email, within 60 seconds of publication.	M
FR-NT-04	Users shall be able to mark notifications read and view a 30-day notification history.	S
FR-NT-05	Notification volume shall respect user preferences (FR-PM-04); emergency alerts override preferences.	S

7.7 Reporting

ID	Requirement	Priority
FR-RP-01	Admins shall be able to view institution-level usage analytics: weekly/monthly active users, notices published, assessments completed, career paths chosen, sessions completed, and commission earned.	S
FR-RP-02	Faculty shall be able to view anonymized, aggregate analytics for their own published content (views, reach).	C
FR-RP-03	Mentors shall be able to view personal earnings reports (sessions, gross, commission, net payout) by period.	S
FR-RP-04	The system shall export admin and mentor reports as CSV.	C

7.8 Administration

ID	Requirement	Priority
FR-AD-01	Admins shall be able to verify/reject mentor applications and suspend mentors for policy violations.	M
FR-AD-02	Admins shall be able to view and act on the dispute queue: inspect the session timeline, chat log, and evidence; rule; and trigger refund or release.	M
FR-AD-03		M

	Admins shall be able to configure the commission rate (bounded to 8-12%) and featured-placement settings.	
FR-AD-04	Admins shall be able to publish/edit/withdraw any notice and broadcast emergency alerts.	M
FR-AD-05	Admins shall be able to suspend or reinstate user accounts and remove abusive content, with every action audit-logged.	M
FR-AD-06	The system shall maintain an immutable audit log of all dispute resolutions, refunds, releases, and administrative overrides.	M

7.9 Error Handling

ID	Requirement	Priority
FR-EH-01	All API errors shall return a consistent JSON error envelope (code, message, trace id) and the UI shall render user-readable messages, never stack traces.	M
FR-EH-02	Payment failures shall be handled transactionally: if bKash capture fails or the callback is not confirmed, the session shall remain in Requested state and no funds shall be recorded as escrowed.	M
FR-EH-03	Idempotent payment callbacks: repeated bKash IPN notifications for the same payment shall not double-record funds or trigger duplicate releases.	M
FR-EH-04	Form validation errors shall be displayed inline per field; the system shall preserve user input on failed submissions.	M
FR-EH-05	The system shall degrade gracefully when external APIs (maps/weather) are unavailable, hiding affected widgets rather than failing the page.	S

Chapter 8 — Use Cases

8.1 Use Case Diagram

Figure 8.1 models the primary use cases of the Campus Guide system and their associations with the four actor groups. Students and faculty interact from the left of the boundary; mentors and administrators from the right.

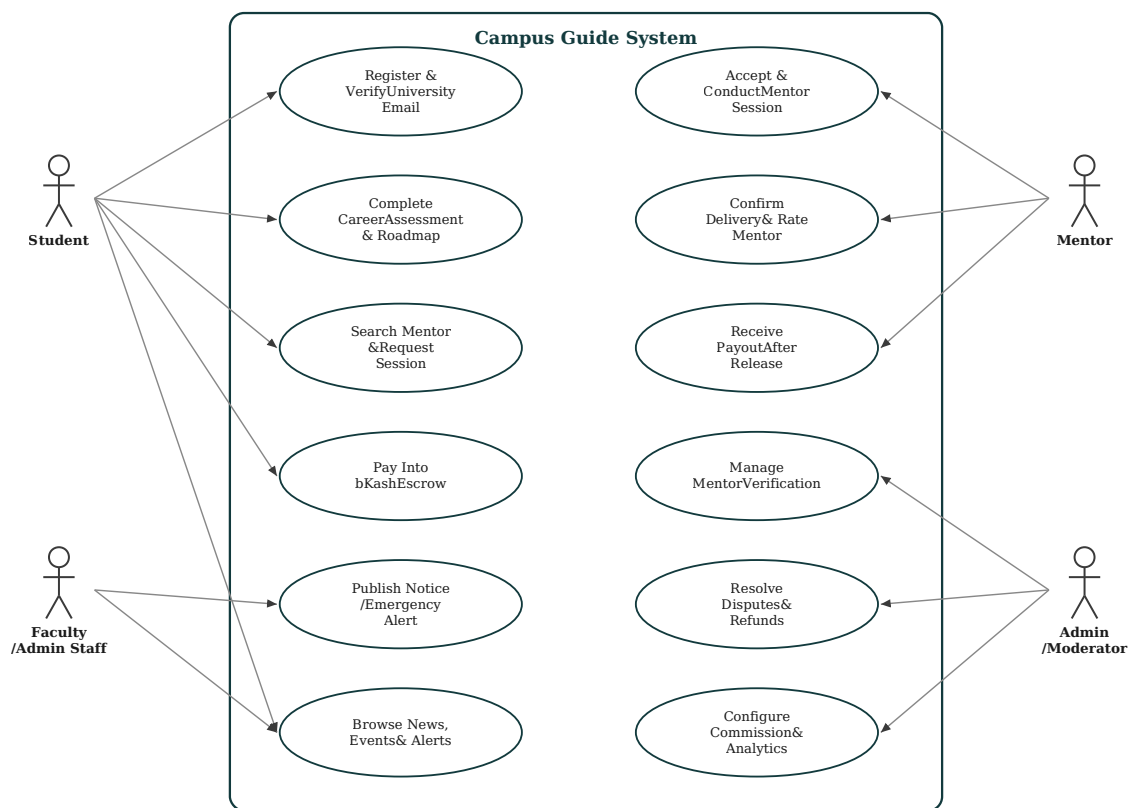


Figure 8.1 — Use case diagram of the Campus Guide platform

8.2 Use Case List

Table 8.1 — Use case list

ID	Use Case	Primary Actor(s)	Brief Description
UC-01		Student, Mentor	

	Register & Verify University Email		Sign up with @uiu.ac.bd email and activate via emailed link.
UC-02	Browse News, Events & Alerts	Student, Faculty	Consume the campus feed with filters and search.
UC-03	Publish Notice / Emergency Alert	Faculty, Admin	Create, schedule, and broadcast official content.
UC-04	Complete Career Assessment & View Roadmap	Student	State interests and job preference; view matched career and roadmap.
UC-05	Search Mentor & Request Session	Student (seeker)	Find a mentor by course code and request a slot.
UC-06	Pay Into bKash Escrow	Student (seeker)	Authorize bKash payment held in escrow for the session.
UC-07	Accept & Conduct Mentor Session	Mentor	Accept request, deliver help, mark delivered.
UC-08	Confirm Delivery & Rate Mentor	Student (seeker)	Confirm completion and submit a rating.
UC-09	Raise & Resolve Dispute	Student, Mentor, Admin	Freeze escrow and obtain an admin ruling with refund or release.
UC-10	Manage Mentor Verification	Admin	Approve/reject mentor applications; suspend mentors.
UC-11	Moderate Content & Accounts	Admin	Remove abusive content, suspend/reinstate accounts, audit log.
UC-12	Configure Commission & View Analytics	Admin	Set commission rate; monitor usage and revenue dashboards.

8.3 Detailed Use Cases

Table 8.2 — Detailed use case UC-01: Register & Verify University Email

Use Case ID / Name	UC-01 — Register & Verify University Email
Primary Actor	Student or prospective mentor
Preconditions	Actor possesses an active @uiu.ac.bd email account.
Postconditions	Account exists with role student, email verified; verification token consumed.
Basic Flow	1. Actor opens the registration page. 2. System requests name, @uiu.ac.bd email, password. 3. System validates inputs (FR-UM-01, FR-EH-04). 4. System creates an unverified account and emails a time-limited verification link. 5. Actor opens the link within 24h. 6. System activates the account and redirects to the role dashboard (FR-UM-06).
Alternate Flows	A1 — Email already registered: system rejects with a sign-in/reset prompt (FR-UM-05). A2 — Link expired: system offers re-send; account remains unverified.

Exceptions	E1 — SMTP failure: system shows a retry state and logs the incident for the admin queue.
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Table 8.3 — Detailed use case UC-04: Complete Career Assessment & View Roadmap

Use Case ID / Name	UC-04 — Complete Career Assessment & View Roadmap
Primary Actor	Student
Preconditions	Actor is logged in (UC-01).
Postconditions	Assessment stored; ranked career matches computed; roadmap saved for the chosen path.
Basic Flow	1. Student opens Career Path Finder. 2. System presents the guided assessment (subjects/topics enjoyed, target job preference). 3. Student submits answers. 4. Matching engine scores the curated career catalogue (FR-CP-02). 5. System displays ranked best-fit careers with match rationale. 6. Student selects a career. 7. System generates and displays the personalized roadmap: courses, certifications, clubs, mentor-session suggestions (FR-CP-03). 8. Student saves the path and begins tracking steps (FR-CP-04).
Alternate Flows	A1 — Re-assessment: previous results preserved for comparison (FR-CP-05). A2 — Changed preferences: roadmap recalculated (FR-CP-06).
Exceptions	E1 — Catalogue unavailable: system shows a friendly error and preserves draft answers.

Table 8.4 — Detailed use case UC-05: Search Mentor & Request Session

Use Case ID / Name	UC-05 — Search Mentor & Request Session
Primary Actor	Student (seeker)
Preconditions	Seeker logged in; verified mentors exist for the course code.
Postconditions	MentorSession created in state Requested; mentor notified.
Basic Flow	1. Seeker searches by course code (FR-MH-02). 2. System returns ranked mentors with rating, completed count, fee, and availability (FR-PM-05). 3. Seeker selects a mentor and a time slot and duration. 4. System computes the fee and shows an order summary. 5. Seeker confirms the request (FR-MH-03). 6. System creates the session in Requested state and notifies the mentor (FR-NT-01).
Alternate Flows	A1 — No mentors found: system suggests adjacent course codes or an alert-me option.
Exceptions	E1 — Concurrent slot conflict: system re-validates availability at confirmation and offers alternative slots.

Table 8.5 — Detailed use case UC-06: Pay Into bKash Escrow

Use Case ID / Name	UC-06 — Pay Into bKash Escrow
Primary Actor	Student (seeker)

Preconditions	A session request exists (UC-05); bKash merchant service reachable.
Postconditions	EscrowTransaction in state Funded; session moves to Accepted on mentor acceptance.
Basic Flow	1. From the request confirmation the seeker chooses bKash payment. 2. System calls the bKash Checkout/Tokenized API to create a payment for the computed amount (FR-MH-04). 3. Seeker completes bKash authorization (OTP/PIN) on bKash's flow. 4. bKash confirms via callback/IPN. 5. System verifies the callback signature and amount, records the EscrowTransaction as Funded, and notifies both parties.
Alternate Flows	A1 — Payment cancelled or failed: session remains Requested with no funds recorded (FR-EH-02); seeker may retry.
Exceptions	E1 — Duplicate IPN: idempotency key prevents double recording (FR-EH-03). E2 — bKash outage: system polls status on a schedule and reconciles before expiring the request.

Table 8.6 — Detailed use case UC-07: Accept & Conduct Mentor Session

Use Case ID / Name	UC-07 — Accept & Conduct Mentor Session
Primary Actor	Mentor
Preconditions	Mentor is verified; a funded session request exists in the mentor's queue.
Postconditions	Session transitions Requested → Accepted → In Progress → Completed (delivered).
Basic Flow	1. Mentor reviews the request (course, topic, time, seeker rating). 2. Mentor accepts within 48h (FR-MH-05). 3. Mentor conducts the session in person or online as described. 4. Mentor marks the session delivered with a short summary; system notifies the seeker to confirm (FR-MH-06).
Alternate Flows	A1 — Mentor declines or times out: full automatic refund; session becomes Declined/Expired (FR-MH-05).
Exceptions	E1 — Seeker no-show: mentor may mark no-show; after 30 minutes the session follows the dispute path (UC-09).

Table 8.7 — Detailed use case UC-08: Confirm Delivery & Rate Mentor

Use Case ID / Name	UC-08 — Confirm Delivery & Rate Mentor
Primary Actor	Student (seeker)
Preconditions	Session is in Completed (awaiting confirm) state.
Postconditions	Escrow released to mentor minus commission; rating published on the mentor profile.
Basic Flow	1. Seeker opens the confirmation prompt. 2. Seeker confirms the session happened as described and submits a 1-5 rating with optional comment (FR-MH-06). 3. System releases escrowed funds to the mentor minus the configured commission, records the payout, and updates the mentor's completed count and rating (FR-MH-07).

Alternate Flows	A1 — No action within 48h: auto-confirm with a neutral system rating (FR-MH-10).
Exceptions	E1 — Release failure at bKash: funds remain escrowed; case enters the admin queue for manual release with retries.

Table 8.8 — Detailed use case UC-09: Raise & Resolve Dispute

Use Case ID / Name	UC-09 — Raise & Resolve Dispute
Primary Actor	Student or Mentor; resolved by Admin
Preconditions	A funded session exists; dispute raised before seeker confirmation.
Postconditions	Escrow either refunded (fully/partially) or released; resolution written to the audit log.
Basic Flow	1. Affected party opens the session and raises a dispute with reason and evidence (FR-MH-08). 2. System freezes the escrowed funds and creates a dispute case. 3. Admin inspects the session timeline, chat log, and evidence in the dispute panel (FR-AD-02). 4. Admin rules: refund seeker, partial refund, or release to mentor (FR-MH-09). 5. System executes the ruling at bKash, updates all parties, and writes the audit record (FR-AD-06).
Alternate Flows	A1 — Both parties cancel amicably before delivery: mutual cancellation triggers automatic full refund.
Exceptions	E1 — Ruling cannot execute at bKash (technical failure): case stays open with manual retry and admin notification.

UC-02, UC-03, UC-10, UC-11 and UC-12 follow the same template and are summarized in Table 8.1; their full flows appear in the use-case repository accompanying the design package.

8.4 Use Case Relationships

- **«include»** — UC-05 (Search & Request Session) includes UC-06 (Pay Into bKash Escrow): every confirmed request must fund escrow before acceptance. UC-08 includes the payout-release step of UC-06's inverse operation (release).
- **«extend»** — UC-09 (Raise & Resolve Dispute) extends UC-06/UC-07/UC-08: it can occur only before confirmation and freezes the otherwise-standard money flow. UC-03's emergency alert extends normal notice publication with broadcast semantics.
- **Generalization** — every actor generalizes to Verified UIU User (UC-01 is the shared entry use case); the Student actor specializes into Seeker behavior when paying, and the Mentor actor specializes from the Student actor (a mentor is a student with an approved mentor profile).

Chapter 9 — External Interface Requirements

9.1 User Interfaces

The user interface shall be a responsive, mobile-first web application rendered server-side by Next.js. Design principles: single verified login, three clearly separated pillars (Info Hub, Career Finder, Mentor Help), role-based navigation, and a calm visual style with WCAG 2.1 AA contrast. Key screens are: home feed (notices/events/alerts with filters and traffic/weather widgets), notice detail, academic calendar, career assessment wizard, career match and roadmap pages, mentor search and mentor profile, session booking and payment confirmation, session history and rating dialogs, dispute form, mentor earnings page, and admin dashboards (verification queue, dispute panel, analytics, commission settings, notice publisher). Wireframe inventory: Appendix E.

9.2 Hardware Interfaces

The platform runs in the browser and defines no direct hardware interfaces. Supported client devices are desktops, laptops, tablets, and smartphones with evergreen browsers (Section 2.4). Cameras/microphones may be used by third-party meeting tools when a mentor session is held online, but the platform only stores links, not media. Server hardware is fully managed by Vercel and the database providers; no institution-owned hardware is used.

9.3 Software Interfaces

Table 9.1 — Software interfaces

System	Interface Mechanism	Purpose / Notes
PostgreSQL 16	Prisma ORM over TLS	Transactional data: users, sessions, escrow, ratings, disputes, audit.
MongoDB Atlas	Mongoose ODM	Document data: notices, events, mentor profiles, chat threads, notifications.
Redis 7	Client library (Upstash REST or equivalent)	Session cache, rate limiting, dashboard caching.
bKash Merchant API	REST (Checkout / Tokenized Checkout), sandbox + production	Payment create, execute, capture; query payment status; release/transfer to mentor; IPN callback verification.
Google Maps JavaScript API	Client-side JS SDK	Campus map, live traffic layer.
OpenWeather API	REST (JSON)	Current weather and short forecast near campus.
Email (SMTP / transactional service)	SMTP relay	Verification, resets, booking lifecycle, dispute outcomes, emergency alerts.

9.4 Communication Interfaces

- All client-server communication over HTTPS/TLS 1.3; HSTS enabled; no plaintext endpoints.
- REST/JSON for all API routes, with consistent error envelopes (FR-EH-01).
- bKash callbacks received via server-side webhook (IPN) with signature verification and idempotent processing (FR-EH-03).
- Session chat: stored messages in MongoDB with polling or server-sent events; no media relay in this release.
- Email notifications via SMTP with SPF/DKIM-aligned sending domain.

9.5 API Requirements

- All API routes are prefixed **/api** and versioned as **/api/v1** once stabilized; the pilot runs on the current route set documented in Appendix F.
- Authenticated routes require a valid session cookie; admin routes additionally require the admin role (RBAC middleware).
- Payment-related routes are idempotent by idempotency key; replayed requests return the original result.
- Rate limits: search and messaging are limited per user per minute (NFR-SE-08); 429 responses include a Retry-After header.
- Requests are validated server-side with schema-based validation; unknown fields are rejected.

Chapter 10 — Non-Functional Requirements

10.1 Performance

ID	Requirement / Target	Metric / Verification
NFR-PE-01	Server-rendered pages shall load in ≤ 2.0 s (P95) on a 4G connection.	Load testing, Lighthouse
NFR-PE-02	API route responses shall return in ≤ 500 ms (P95) for read endpoints.	Load testing
NFR-PE-03	Mentor search shall return ≤ 2 s (P95) for the pilot dataset.	Test TC-PE-1
NFR-PE-04	The platform shall support at least 500 concurrent users with no error rate above 1% during pilot peaks (exam weeks).	Stress test

10.2 Security

ID	Requirement / Target	Metric / Verification
NFR-SE-01	The platform shall conform to OWASP ASVS Level 2; the OWASP Top 10 (2021) shall be addressed in design and verified in testing.	Security test pass, Section 15.7
NFR-SE-02	Passwords shall be hashed with bcrypt (cost ≥ 10) or Argon2id; plaintext or reversible storage is prohibited.	Code audit
NFR-SE-03	All transport shall use TLS 1.3 with HSTS; cookies shall be Secure and HttpOnly.	Scan / config audit
NFR-SE-04	Authorization shall be enforced server-side on every route (RBAC: student / mentor / faculty / admin); client-side checks are cosmetic only.	Penetration test
NFR-SE-05	All database access shall use parameterized queries via Prisma/ Mongoose; raw string SQL is prohibited.	Code audit
NFR-SE-06	State-changing forms shall carry per-session CSRF tokens.	Automated checks
NFR-SE-07	bKash callbacks shall be signature-verified and processed idempotently.	Integration test
NFR-SE-08	Rate limiting shall protect login, search, messaging, and listing endpoints against abuse and scraping.	Load/abuse test
NFR-SE-09	Audit logs (disputes, refunds, releases, admin actions) shall be append-only and retained ≥ 2 years.	Schema + ops audit

10.3 Usability

ID	Requirement / Target	Metric / Verification
NFR-US-01	A first-time student shall reach the feed and complete onboarding in ≤ 5 minutes without documentation.	UAT observation

NFR-US-02	Core actions (find a notice, book a mentor) shall be reachable in ≤ 3 clicks from the dashboard.	UAT observation
NFR-US-03	The booking and escrow flow shall state clearly at each step what is charged, what is held, and when it is released.	UAT feedback
NFR-US-04	All user-visible error messages shall be in plain language and suggest a next action.	Review

10.4 Reliability

ID	Requirement / Target	Metric / Verification
NFR-RE-01	Failed requests shall be retried with backoff where safe; money operations shall retry only idempotently.	Integration test
NFR-RE-02	Databases shall be backed up daily with point-in-time recovery for PostgreSQL; restore drills shall be performed monthly during pilot.	Ops runbook
NFR-RE-03	Payment state shall be reconciled with bKash status queries on a schedule, so no funded session can be silently lost.	Reconciliation report

10.5 Availability

ID	Requirement / Target	Metric / Verification
NFR-AV-01	The platform shall target 99.5% monthly availability during the pilot semester.	Uptime monitoring
NFR-AV-02	Planned maintenance shall occur off-peak with ≥ 24 h in-app notice.	Ops policy

10.6 Scalability

ID	Requirement / Target	Metric / Verification
NFR-SC-01	Stateless Next.js deployment shall scale horizontally on the managed platform without code change.	Architecture review
NFR-SC-02	The data model shall be tenant-scoped (institution id on core tables) so that a second campus can be added without schema migration.	Design review
NFR-SC-03	Feed and search endpoints shall be cache-friendly (Redis) to absorb exam-week spikes.	Load test

10.7 Maintainability

ID	Requirement / Target	Metric / Verification
NFR-MA-01		CI report

	The codebase shall be TypeScript with a target of $\geq 70\%$ unit-test coverage on business logic.	
NFR-MA-02	Business rules (commission bounds, timeout windows, assessment weights) shall be externalized in configuration, not hard-coded.	Code audit
NFR-MA-03	CI shall run build, lint, and tests on every pull request; deployments are automated and reversible.	CI/CD audit

10.8 Portability

ID	Requirement / Target	Metric / Verification
NFR-PO-01	The UI shall function on the latest two versions of Chrome, Edge, Firefox, and Safari, desktop and mobile.	Cross-browser test matrix
NFR-PO-02	The application shall be portable across Node 20+ hosts; only managed-service credentials bind it to a provider.	Config audit

10.9 Accessibility

ID	Requirement / Target	Metric / Verification
NFR-AC-01	The platform shall meet WCAG 2.1 Level AA: contrast, keyboard operability, focus order, alt text, and semantic landmarks.	Accessibility audit, Section 15
NFR-AC-02	All interactive flows (booking, payment confirmation, dispute) shall be fully keyboard-navigable with visible focus states.	Manual + automated audit

10.10 Privacy

ID	Requirement / Target	Metric / Verification
NFR-PR-01	The platform shall collect only the personal data needed for its functions (identity via university email, profile fields, transaction records); no sensitive special-category data shall be stored.	Privacy review
NFR-PR-02	On account closure, personal identifiers shall be anonymized within 30 days; transaction records required for financial audit shall be retained in anonymized form.	Retention runbook
NFR-PR-03	Personal contact details shall be exchanged only after a session relationship exists; the platform shall not expose emails or phone numbers on public profiles.	Design review
NFR-PR-04	Analytics dashboards shall present only aggregate, non-identifying statistics to faculty and admins.	Review

Chapter 11 — System Models and Data Design

11.1 System Architecture

Campus Guide uses a layered architecture executed on a single Next.js codebase (Figure 11.1). The client layer is the browser (SSR pages plus progressive enhancement). The application layer contains the App Router pages, API route handlers, and the NextAuth/RBAC middleware that guards every route. The service layer implements the six business services (Info Hub, Career Matcher, Booking & Scheduling, Escrow Orchestration, Notification, Analytics). The data layer spans PostgreSQL (transactional truth), MongoDB (document data), Redis (cache and rate limits), and object storage for avatars and verification documents. External systems (bKash, Maps, OpenWeather, email) are reached over REST from the service layer.

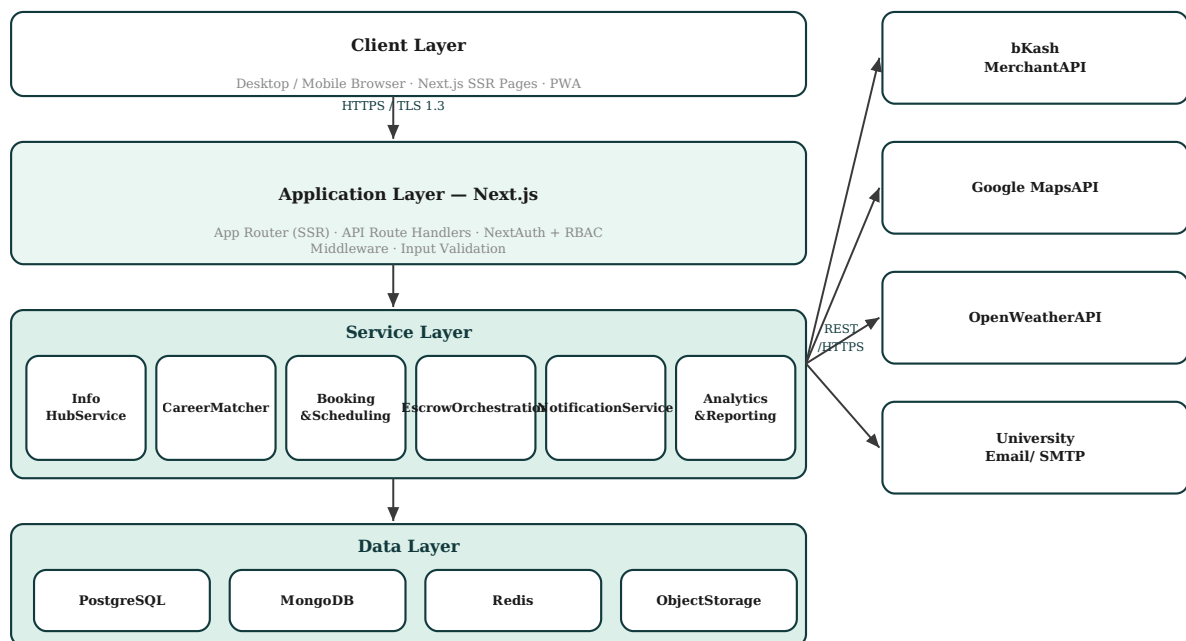


Figure 11.1 — Layered system architecture of Campus Guide

11.2 Context Diagram

Figure 11.2 shows the system boundary. Students supply profiles, assessments, and booking requests and receive notices, career recommendations, and escrow receipts. Mentors supply availability and session delivery and receive requests and payouts. Faculty and administrators publish content and moderation actions and receive analytics and dispute cases. bKash receives fund/release instructions and returns payment confirmations. The university email/IT directory verifies identity; Maps and OpenWeather return traffic and weather data.

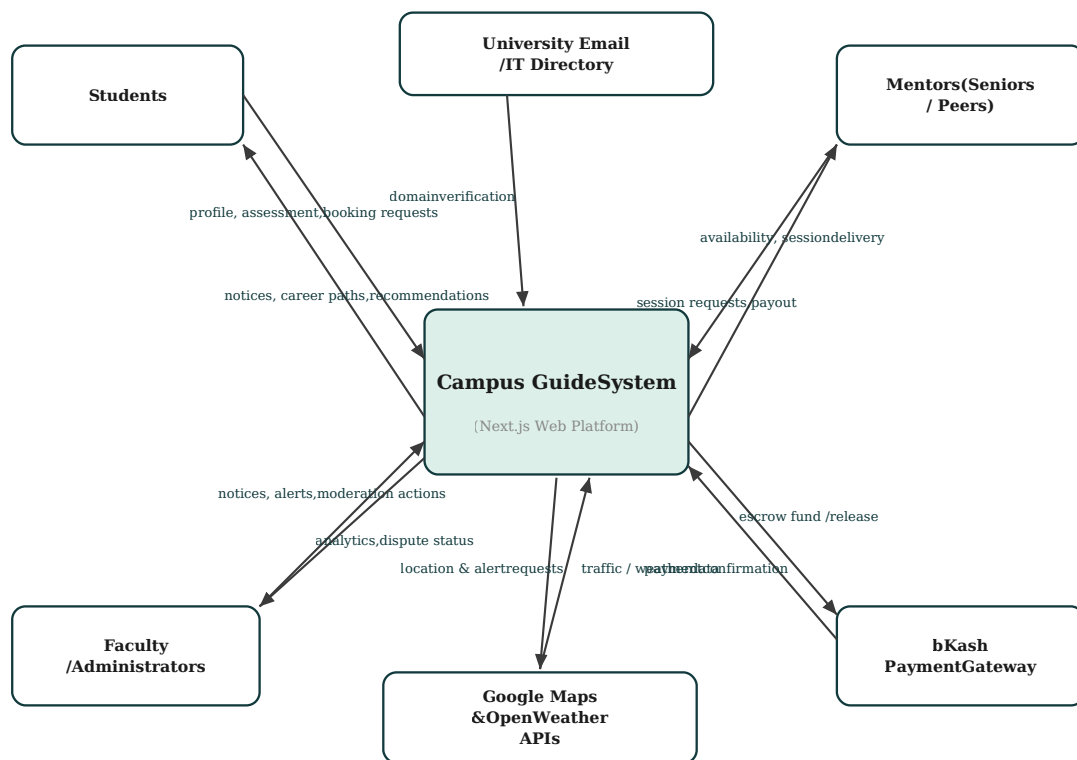


Figure 11.2 — Context diagram of the Campus Guide system

11.3 Data Flow Diagram

The level-0 DFD (Figure 11.3) decomposes the system into six processes with three data stores.

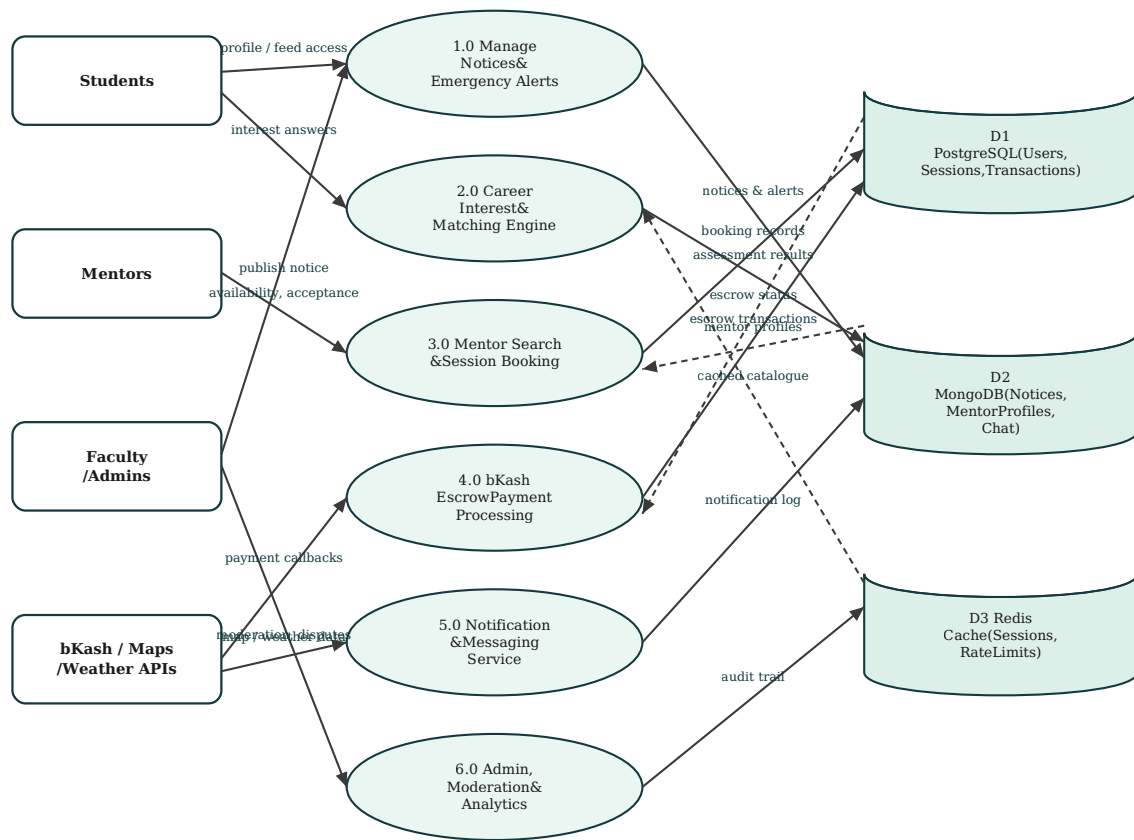


Figure 11.3 — Level-0 data flow diagram

Table 11.1 — DFD process descriptions

Process	Name	Description
1.0	Manage Notices & Emergency Alerts	Accepts publications from faculty/admins, serves the filtered feed to students, and broadcasts emergency alerts.
2.0	Career Interest & Matching Engine	Collects interest/preference assessments, scores them against the career catalogue, and persists results and roadmaps.
3.0	Mentor Search & Session Booking	Serves course-code mentor search, validates availability, and creates session requests and acceptances.
4.0	bKash Escrow Payment Processing	Creates and captures bKash payments into escrow, verifies callbacks idempotently, and executes release or refund rulings.
5.0	Notification & Messaging Service	Dispatches in-app and email notifications for all lifecycle events; stores session chat threads.

6.0	Admin, Moderation & Analytics	Handles mentor verification, dispute resolution, content moderation, commission configuration, and usage analytics; writes the audit trail.
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11.4 UML Models

The static structure is captured in the class diagram (Figure 11.4). Behavioral models — the booking/escrow sequence, the session/escrow state machine, and the deployment view — appear in Appendix D (Figures D.1–D.3) and are normative parts of this specification.

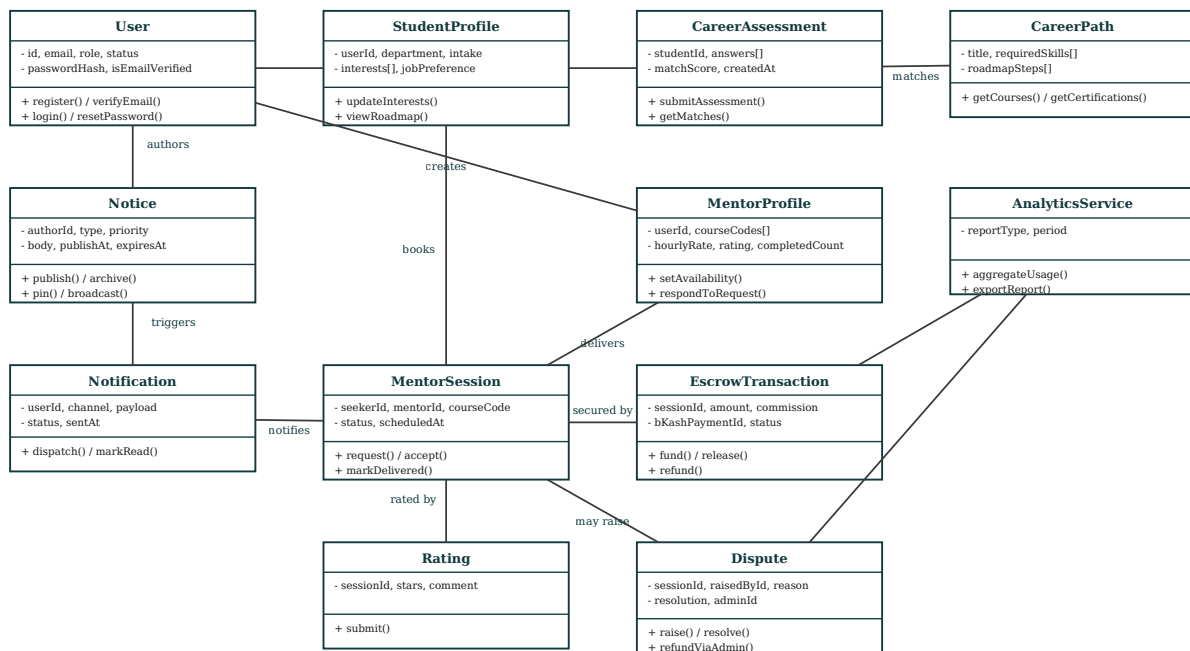


Figure 11.4 — Class diagram of the core domain

11.5 ER Diagram

Figure 11.5 presents the entity-relationship model of the transactional core (PostgreSQL), with the document collections of MongoDB noted in Section 11.6.

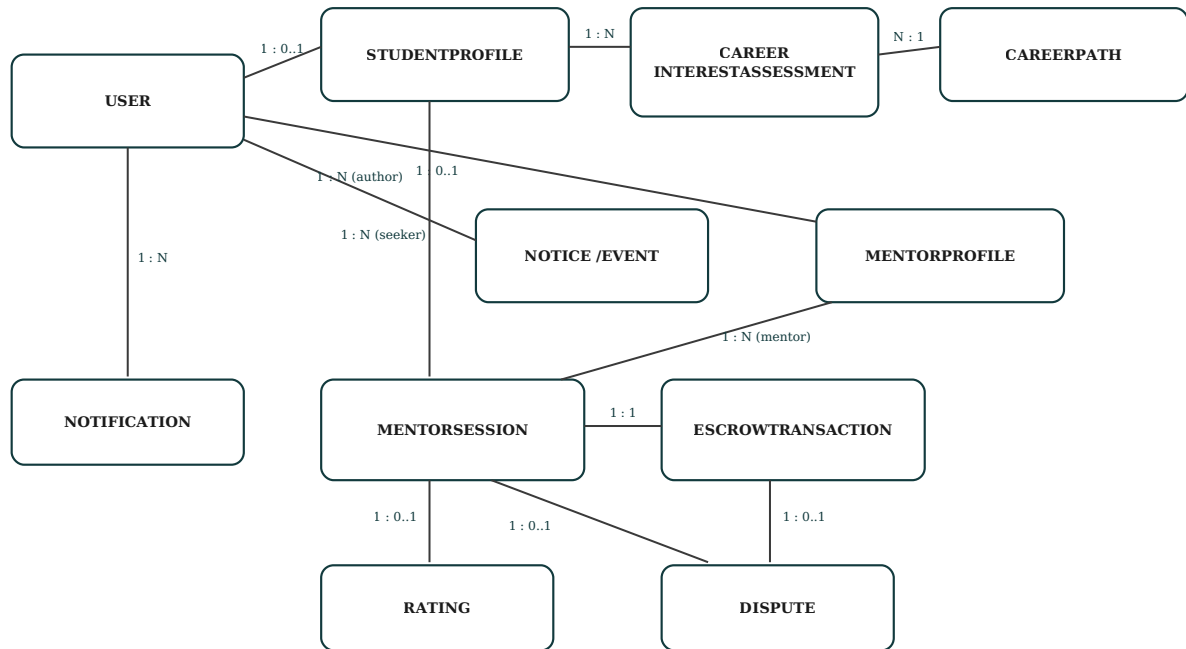


Figure 11.5 — Entity-relationship diagram of the transactional core

Table 11.2 — Entity relationships

Relationship	Cardinality	Meaning
User — StudentProfile	1 : 0..1	Every student account extends the user with academic attributes.
User — MentorProfile	1 : 0..1	A student may hold one mentor profile (after verification).
User — Notice	1 : N	Faculty/admins author notices and events.
StudentProfile — CareerInterest	1 : N	A student may take multiple assessments over time.
CareerInterest — CareerPath	N : 1	Each assessment matches to one selected career path.

StudentProfile — MentorSession (seeker)	1 : N	Sessions booked by the student.
MentorProfile — MentorSession (mentor)	1 : N	Sessions delivered by the mentor.
MentorSession — EscrowTransaction	1 : 1	Each paid session has exactly one escrow record.
MentorSession — Rating	1 : 0..1	Rating written on confirmation.
MentorSession — Dispute	1 : 0..1	At most one dispute per session.

11.6 Database Design

Data is split by shape and access pattern. PostgreSQL is the transactional system of record for anything involving identity, money, or state machines. MongoDB stores the publication/feed documents and chat threads whose shapes evolve independently. Redis holds only disposable cache and counters.

Table 11.3 — PostgreSQL tables (system of record)

Table	Purpose	Key Columns
users	Identity and credentials	id, email, password_hash, role, status, email_verified_at, institution_id
student_profiles	Academic attributes	user_id (FK), department, intake, interests[], job_preference
career_assessments	Assessment submissions	id, student_id (FK), answers (jsonb), matched_path_id, match_score, created_at
career_paths	Curated catalogue	id, title, description, required_skills (jsonb), roadmap_template (jsonb), active
mentor_profiles	Mentor offering & track record	user_id (FK), course_codes[], hourly_rate, verification_status, rating_avg, completed_count, payout_msisdn
mentor_sessions	Booking state machine	id, seeker_id, mentor_id, course_code, topic, scheduled_at, duration_min, status, delivered_at, confirmed_at
escrow_transactions	Money in escrow	id, session_id (FK), amount, commission_pct, bkash_payment_id, status, funded_at, released_at, refunded_at
ratings	Session feedback	id, session_id (FK), stars, comment, by_role, created_at

disputes	Dispute cases	id, session_id (FK), raised_by_id, reason, evidence_url, status, resolution, resolved_by_id, resolved_at
audit_logs	Append-only admin actions	id, actor_id, action, entity, entity_id, detail (jsonb), created_at

Table 11.4 — MongoDB collections (documents)

Collection	Purpose	Notable Fields
notices	Feed content	type, category, priority, title, body, authorRef, audience[], publishAt, expiresAt, isEmergency
events	Calendar items (seminars, exams windows)	title, kind, startsAt, endsAt, location, department, source
mentor_public	Denormalized searchable mentor cards	userId, name, dept, courseCodes[], rate, rating, completedCount, availability
chat_threads	Session messaging	sessionId, participants[], messages[{sender, body, at}]
notifications	Notification history	userId, kind, payload, read, createdAt

11.7 Data Dictionary

Table 11.5 — Data dictionary (selected elements)

Element	Type / Format	Description & Validation
email	string, RFC 5322	Must end with @uiu.ac.bd; unique per institution.
password	string	≥ 8 chars with letters and digits; stored as bcrypt/Argon2 hash only.
role	enum	student mentor faculty admin (mentor is a capability on a student account).
course_code	string	Pattern: 3-4 letters + '-' + 4 digits (e.g., CSE-1109); validated against UIU catalogue list.
hourly_rate	integer BDT	$1 \leq \text{rate} \leq 2000$ per hour; mentor-set.
amount (escrow)	integer BDT	$\text{hourly_rate} \times \text{ceil}(\text{duration_min}/60)$; $1 \leq \text{amount} \leq 5,000$.
commission_pct	decimal	Bounded 8.0-12.0; admin-configurable (FR-AD-03).
session.status	enum	Requested Accepted EscrowFunded InProgress Completed Confirmed Disputed Refunded Declined (see Figure D.2).
escrow.status	enum	Initiated Funded Frozen Released Refunded.

stars	integer	1-5; one rating per session per side.
match_score	decimal 0-100	Weighted score of interest/preference overlap with catalogue path.

11.8 Data Validation

- All inputs are schema-validated server-side at the API boundary; validation failures return field-level errors (FR-EH-04).
- Monetary amounts are integers (BDT, no fractions); no float arithmetic on money.
- State transitions of sessions/escrow are guarded by a server-side state machine; illegal transitions are rejected and logged (Figure D.2).
- Cross-field invariants: a session cannot be EscrowFunded without a verified bKash payment id; Confirmed requires a prior Completed; a dispute cannot be raised after confirmation.
- Referential integrity is enforced by PostgreSQL foreign keys; document references from MongoDB carry the owning user id for authorization checks.

Chapter 12 — Constraints, Assumptions, Dependencies

12.1 Constraints

The binding constraints are consolidated in Section 2.5 (C-1 through C-8): the fixed Next.js + PostgreSQL + MongoDB stack, bKash-only payments, @uiu.ac.bd-verified signup, managed hosting, student-project budget, the 16-week schedule, OWASP ASVS L2 / WCAG 2.1 AA compliance targets, and the merchant-account escrow model that forbids stored wallet balances.

12.2 Assumptions

- The university grants permission for email-domain verification and institutional branding (A-1).
- A pilot cohort of mentors and seekers volunteers in the first semester (A-2).
- Faculty publish notices through the platform at least as often as through legacy channels (A-3).
- Assessment honesty: students answer the career assessment sincerely; results are advisory, not binding.
- Pilot volume stays within the managed-service free or low tiers until commission revenue justifies upgrades.

12.3 Dependencies

Table 12.1 — External dependencies

Dependency	Providing Party	Needed For
bKash Merchant API + sandbox	bKash (merchant partner)	Escrow capture, release, refund, IPN (FR-MH-04/07/09)
University email / SMTP relay	UIU IT	Signup verification, resets, notifications (FR-UM-02, FR-NT-02)
Google Maps API key	Google Maps Platform	Traffic layer near campus (FR-CI-05)
OpenWeather API key	OpenWeather	Weather widgets (FR-CI-05)
Vercel / Neon / Atlas managed hosting	SaaS providers	Application and data hosting (D-4)

12.4 Dependency Failure Impact

Table 12.2 — Dependency failure modes and mitigations

Dependency	Failure Mode	Impact	Mitigation
bKash API	Outage or sandbox expiry	No new escrow funding; existing funds unaffected	Queue requests with pay-later; reconciliation job; status page notice
University SMTP	Relay blocked / delayed	Signups and resets stall	Fallback transactional email provider; admin manual-verify path
Maps / Weather APIs	Quota exceeded	Widgets degrade	Cache responses 15 min; hide widgets gracefully (FR-EH-05)
Managed hosting	Regional outage	Platform unavailable	Provider status page monitored; SLA targets (NFR-AV-01); daily backups elsewhere
University cooperation	Email verification policy change	New registrations blocked	Offline verification queue by admins; direct dialogue with IT

Chapter 13 — Risk Analysis

13.1 Risk Identification

Risks were identified in workshops with the supervisor and pilot stakeholders, scored on likelihood (L) and impact (I) as High/Medium/Low, with exposure = $L \times I$.

Table 13.1 — Risk register

ID	Risk	Cat.	L	I	Exposure	Mitigation Ref.
R-01	bKash integration issues (callbacks, sandbox drift)	Tech	M	H	High	M-01
R-02	Cold-start: too few mentors for credible search	Oper.	H	H	Critical	M-02
R-03	Payment dispute abuse / fake claims	Sec	M	H	High	M-03
R-04	Data breach or credential compromise	Sec	L	H	Medium	M-04
R-05	Team capacity during exams (bus factor)	Sched	M	M	Medium	M-05
R-06	Faculty adoption lag (stale feed)	Oper	M	M	Medium	M-06
R-07	External API quota exhaustion (maps/weather)	Tech	M	L	Low	M-07
R-08	Scope creep beyond the 16-week plan	Sched	M	M	Medium	M-08

13.2 Technical Risks

The dominant technical risk (R-01) is the bKash escrow orchestration: signature verification, idempotent IPN processing, and state reconciliation. The design answers with a dedicated Escrow Orchestration service, idempotency keys (FR-EH-03), and a scheduled reconciliation job (NFR-RE-03), all tested first against the sandbox. R-07 is mitigated by response caching and graceful degradation (FR-EH-05).

13.3 Security Risks

Primary security risks are payment abuse (R-03) and account/credential compromise (R-04). Controls: university-email verification, RBAC on every route, ASVS L2 practices (NFR-SE-01..09), append-only audit logs for all money events, dispute evidence trails, and rate limiting. Dispute abuse is further discouraged by a published policy: repeated bad-faith disputes lead to account suspension (FR-AD-05).

13.4 Operational Risks

R-02 (cold start) is the platform's biggest existential risk: a mentor search with empty results kills trust at first use. Mitigation: seed 15-20 verified mentors across high-demand

courses before launch, guarantee first-month visibility, and market through department champions. R-06 is mitigated by making publication easier than the Facebook group and by launch-partner agreements with two departments.

13.5 Schedule Risks

R-05 and R-08 threaten the 16-week schedule. Mitigation: the 4-week MVP core is sacrosanct (Section 14.3); S- and C-priority requirements are explicitly droppable; and a two-person-review rule keeps knowledge shared. Milestone reviews every two weeks surface slippage early.

13.6 Risk Mitigation

Table 13.2 — Mitigation plan

Ref	Mitigation	Owner
M-01	Sandbox-first integration, callback replay tests, reconciliation job, bKash escalation contact	Backend lead
M-02	Seed mentor cohort ≥ 15 before pilot; guaranteed featured placement first month	Ops lead
M-03	Dispute policy with evidence requirements, audit trail, suspension for bad-faith disputes	Admin team
M-04	ASVS L2 checklist, dependency scanning in CI, secrets management, monthly patch review	Security lead
M-05	Cross-review rule, documented runbooks, sprint buffer weeks	Project lead
M-06	Department launch partners; publication easier than legacy channels	Ops lead
M-07	15-minute response cache; widget degradation	Backend lead
M-08	Change control via RTM; MoSCoW-based descoping at milestone reviews	Project lead

13.7 Risk Monitoring

The risk register is reviewed at each bi-weekly milestone: exposure is re-scored, new risks are added, and mitigations are checked for effect. Automated monitoring supports the review: uptime and error-rate alerts (availability), reconciliation reports (payments), moderation queue depth (abuse), and burndown against the plan (schedule). A risk that

triggers its threshold (e.g., mentors active < 10, or any payment reconciliation mismatch) opens an incident with a named owner and a deadline.

Chapter 14 — Project Plan and Estimation

14.1 Methodology

The project follows an **iterative-incremental (Agile) methodology** in eight two-week sprints within the 16-week schedule. Requirements are fixed in this SRS; each sprint delivers a demonstrable increment against the milestone table. Practices: shared backlog, sprint demo with the supervisor, definition-of-done including unit tests and a11y checks, and trunk-based development with CI. The proposal's four-week MVP path maps to Sprints 1-2 (foundation), Sprint 3 (career + booking + sandbox pay), Sprint 4 (trust layer), closing with the pilot launch — so a usable product exists by week 8 even if later sprints slip.

14.2 Work Breakdown Structure

Table 14.1 — Work breakdown structure

WBS	Work Package
1. Requirements	1.1 Stakeholder consultation · 1.2 SRS authoring · 1.3 Sign-off & baseline
2. Design	2.1 Architecture & data model · 2.2 API contracts · 2.3 UI/UX wireframes
3. Foundation	3.1 Auth + university email verification · 3.2 RBAC · 3.3 Notice/calendar feed skeleton
4. Info Hub	4.1 Publishing flow · 4.2 Feed & search · 4.3 Emergency alerts · 4.4 Maps/weather widgets
5. Career Path Finder	5.1 Assessment wizard · 5.2 Matching engine · 5.3 Roadmap generation & tracking
6. Mentor Marketplace	6.1 Mentor onboarding/verification · 6.2 Search & booking · 6.3 Session lifecycle & chat
7. Escrow	7.1 bKash sandbox integration · 7.2 Capture/release/refund · 7.3 Disputes & admin panel
8. Quality	8.1 Test suites · 8.2 Security audit · 8.3 Accessibility audit · 8.4 UAT
9. Launch	9.1 Seed cohort · 9.2 Pilot launch · 9.3 Monitoring & stabilization

14.3 Schedule

Table 14.2 — Phase schedule (16 weeks)

Phase	Weeks	Deliverables / Milestone
Requirements & design	W1-W2	SRS sign-off (M1); architecture, ERD, API contracts, wireframes (M2)
Foundation	W3-W4	Auth + email verification live; feed skeleton demo (M3)
Campus Info Hub	W4-W6	Publishing, feed, alerts, calendar, maps/weather (M4)
Career Path Finder	W5-W7	Assessment + matching + roadmap (M5)

Mentor marketplace	W6-W8	Search, booking, session lifecycle (M6)
bKash escrow	W8-W10	Sandbox → production escrow, release & refund (M7)
Trust & admin	W10-W11	Ratings, disputes, admin console, audit log (M8)
Testing & audits	W11-W13	Integration/system tests; security & accessibility audits (M9)
UAT & pilot	W13-W16	UAT with pilot cohort; seed mentors; pilot launch & stabilization (M10)

The Gantt chart in Appendix I (Figure I.1) visualizes this schedule.

14.4 Effort Estimation

Table 14.3 — Effort estimation (person-weeks)

Work Area	Effort	Basis
Requirements & design	4	2 people × 2 weeks
Foundation (auth, RBAC, feed)	4	2 × 2 weeks
Info Hub features	4	3-week span, parallel
Career Path Finder	4	Assessment + engine + roadmap
Mentor marketplace	5	Largest business area
bKash escrow	4	Integration + reconciliation + dispute paths
Admin console & analytics	3	Queues, panel, reports
Testing, audits, UAT, launch	6	Quality gate + pilot
Total	34 person-weeks	≈ 2 full-time equivalents × 16 weeks + support

14.5 Cost Estimation

Development labor is course-funded (no cash cost). The recurring cash cost is the infrastructure in Table 3.2: ≈ 30,000–60,000 BTT for the pilot year, plus bKash merchant transaction fees on session volume. Commission income (8–12% per completed session) is projected to cover running costs from roughly 200 completed sessions per month (Section 3.2).

14.6 Milestones

Table 14.4 — Milestones

ID	Milestone	Week	Acceptance Evidence
M1	SRS signed off & baselined	W2	Stakeholder signatures; RTM baseline
M2	Design package complete	W2	Architecture, ERD, API contracts, wireframes reviewed
M3	Foundation demo	W4	Verified signup/login working on staging

M4	Info Hub feature-complete	W6	Publishing + feed + alerts pass TC suite
M5	Career Path Finder complete	W7	End-to-end assessment → roadmap demo
M6	Mentor marketplace complete	W8	Search → request → accept flow on staging
M7	Escrow live in sandbox & production-configured	W10	Full payment lifecycle with callbacks verified
M8	Trust & admin layer complete	W11	Dispute panel + audit log demoed
M9	Quality gate passed	W13	Test report, security & accessibility audit results
M10	Pilot launch	W16	≥ 15 active mentors; first real sessions completing

14.7 Resource Allocation

Table 14.5 — Team roles

Role	Responsibility	Allocation
Project lead / architect	Planning, architecture, stakeholder liaison, integration decisions	50% overall; heavy W1-2
Backend lead	API routes, data layer, escrow orchestration, reconciliation	Full, W3-W10
Frontend lead	UI implementation, accessibility, design system	Full, W3-W13
QA / security engineer	Test suites, security & accessibility audits, UAT coordination	50%, peaking W11-W13
Operations (admin)	Seed mentors, moderation, dispute handling, launch support	Pilot cohort, from W10
Faculty supervisor	Reviews at milestones; escalation path	Bi-weekly

Chapter 15 — Acceptance Criteria and Test Strategy

15.1 Acceptance Criteria

The release is accepted when the following criteria are demonstrably met:

1. All M-priority functional requirements pass their mapped test cases (traceability: Chapter 16).
2. The full escrow lifecycle (request → pay → deliver → confirm → release, plus decline-refund and dispute-refund paths) executes without manual intervention or money-state anomalies.
3. No payment may be recorded without a verified bKash confirmation; no funds move outside the defined state machine.
4. Performance targets (NFR-PE-01..04) met at 200 concurrent simulated users.
5. Security audit: no High or Critical findings open at sign-off (Section 15.7).
6. Accessibility: WCAG 2.1 AA audit with no blocking findings on core flows.
7. Pilot readiness: ≥ 15 verified mentors seeded and the admin runbook exercised once end-to-end.

15.2 Unit Testing

Unit tests cover business logic in isolation: matching-engine scoring, fee computation, commission bounds, state-machine transitions, validation schemas, and idempotency handling. Target ≥ 70% coverage on business logic (NFR-MA-01); CI blocks merges below the threshold.

15.3 Integration Testing

Integration tests exercise the seams: API route → service → database; escrow service → bKash sandbox (create, execute, callback/IPN, release, refund); notification service → email provider; maps/weather with recorded responses. Payment tests replay captured callbacks to prove idempotency (FR-EH-03).

15.4 System Testing

End-to-end scenarios run on staging against production-like data: a student discovers a notice, completes an assessment, books a mentor, pays through sandbox bKash, attends, confirms, and rates; a second scenario walks a dispute to refund. Cross-browser and responsive checks run per NFR-PO-01.

15.5 Acceptance Testing

UAT with the pilot cohort: 5 students, 3 mentors, 1 faculty publisher, 1 admin. Each runs a scripted task list covering the three pillars; task completion without assistance and post-task feedback are recorded. Sign-off requires all M-priority tasks completed and no critical usability objections.

15.6 Regression Testing

The automated unit/integration suites run on every pull request and nightly on main. Payment-state and dispute flows are mandatory regression scope for any change touching the escrow service. Releases are blocked on a green regression run.

15.7 Security Testing

Security testing executes an OWASP Top 10 checklist against ASVS L2: authentication and session controls, RBAC route probing (horizontal/vertical privilege tests), CSRF, injection scan (dependency + code), rate-limit behavior, callback signature verification, and audit log integrity. Any High/Critical finding blocks release until fixed and retested.

15.8 Performance Testing

Load tests simulate 200 and 500 concurrent users across feed, search, and booking paths; targets are NFR-PE-01..04. Stress tests push to 2× target to observe degradation behavior; soak tests run 4 hours to detect leaks in long-lived connections and caches.

15.9 Test Cases

Representative test cases are listed below; the full suite (60+ cases) is maintained with the test plan. Appendix G expands the payment and dispute cases.

Table 15.1 — Representative test cases

ID	Scenario	Related Req.	Expected Result
TC-01	Register with non-UIU email	FR-UM-01	Registration rejected with clear message; no account created
TC-02	Verify account via email link	FR-UM-02	Account activated; role dashboard shown
TC-03	Login rate limiting	FR-AU-03	6th failed attempt within 15 min is locked out with email unlock
TC-04	Publish emergency alert	FR-CI-02	All users see pinned alert within 60 s; email sent
TC-05	Career assessment produces ranked matches	FR-CP-02	Ranked list with scores and rationale; roadmap generated
TC-06	Search mentor by course code	FR-MH-02	Only verified mentors for that course returned, ranked
TC-07	Successful escrow payment (sandbox)	FR-MH-04	EscrowTransaction = Funded; session can be accepted
TC-08	Failed/cancelled bKash payment	FR-EH-02	No escrow record; session stays Requested; retry offered
TC-09	Duplicate bKash IPN	FR-EH-03	Single escrow record; no duplicate funds or releases
TC-10			

	Mentor declines request → auto-refund	FR-MH-05	Full refund executed; both parties notified; state Declined
TC-11	Confirm & rate → payout release	FR-MH-07	Funds released minus commission; rating visible on profile
TC-12	Dispute freezes escrow	FR-MH-08	Funds Frozen; admin case created; no release possible
TC-13	Admin rules dispute for seeker	FR-MH-09	Refund executed; audit log entry with admin, rationale, amounts
TC-14	Admin sets commission out of bounds	FR-AD-03	Change rejected (must stay within 8-12%)
TC-15	Accessibility check of booking flow	NFR-AC-01/02	Keyboard-operable end-to-end; no blocking AA violations

Chapter 16 — Requirements Traceability Matrix

16.1 Requirement-to-Use-Case Mapping

Table 16.1 — Functional requirements ↔ use cases

Requirement	Use Case(s)	Requirement	Use Case(s)
FR-UM-01..03, 05, 06	UC-01	FR-MH-01..03	UC-05, UC-10
FR-AU-01..05	UC-01, UC-11	FR-MH-04	UC-06
FR-PM-01..05	UC-01, UC-05	FR-MH-05..07, 10	UC-06, UC-07, UC-08
FR-CI-01..04, 06..08	UC-02, UC-03	FR-MH-08, 09	UC-09
FR-CI-05	UC-02	FR-SF-01..04	UC-02, UC-05
FR-CP-01..06	UC-04	FR-NT-01..05	UC-02..09 (cross-cutting)
FR-RP-01..04	UC-12, UC-11	FR-AD-01..06	UC-09..UC-12
FR-EH-01..05	Cross-cutting (all)	—	—

16.2 Requirement-to-Design Mapping

Table 16.2 — Requirements ↔ design elements

Requirement Group	Design Element (Chapter 11 / Appendix D)
User management & authentication (FR-UM, FR-AU)	NextAuth + RBAC middleware (Fig. 11.1); users table; context diagram identity flows (Fig. 11.2)
Campus Info Hub (FR-CI)	Info Hub Service (Fig. 11.1); Process 1.0 and D1/D2 flows (Fig. 11.3); notices/events collections (Table 11.4)
Career Path Finder (FR-CP)	Career Matcher service; Process 2.0 (Fig. 11.3); career_assessments / career_paths tables; class model (Fig. 11.4)
Mentor marketplace (FR-MH-01..03, 05, 06, 10)	Booking & Scheduling service; Process 3.0; mentor_sessions table; sequence diagram (Fig. D.1); state machine (Fig. D.2)
Escrow & disputes (FR-MH-04, 07..09)	Escrow Orchestration service; Process 4.0; escrow_transactions / disputes tables; state machine (Fig. D.2)
Notifications (FR-NT)	Notification Service; Process 5.0; notifications collection
Administration & reporting (FR-AD, FR-RP)	Admin console; Process 6.0; audit_logs table; Analytics service (Fig. 11.1)
Error handling & idempotency (FR-EH)	API error envelope; escrow reconciliation job; guarded state machine (Section 11.8)

16.3 Requirement-to-Test Mapping

Table 16.3 — Requirements ↔ test cases

Requirement	Test Case(s)	Requirement	Test Case(s)
FR-UM-01..02	TC-01, TC-02	FR-MH-04	TC-07, TC-08
FR-AU-03	TC-03	FR-MH-05	TC-10
FR-CI-02	TC-04	FR-MH-07	TC-11
FR-CP-02	TC-05	FR-MH-08..09	TC-12, TC-13
FR-MH-02	TC-06	FR-AD-03	TC-14
FR-EH-02..03	TC-08, TC-09	NFR-AC-01/02	TC-15

16.4 Coverage Analysis

All 64 functional requirements are covered by at least one use case and one design element; all M-priority requirements have at least one mapped test case (15 representative cases shown here; the full suite covers each M requirement individually). NFRs are covered by the dedicated audits and performance/accessibility test passes in Sections 15.7–15.8. Coverage is recomputed automatically from the RTM at every milestone; the release gate (Section 15.1) requires zero uncovered M-priority requirements.

16.5 Change Tracking

Table 16.4 — Change tracking log (baseline 1.0)

Change	Affected Reqs.	Description	Date	Status
CH-001	FR-MH-10	Added 48-hour auto-confirm window to protect mentors from unresponsive seekers	2026-09-10	Approved
CH-002	FR-AD-03	Commission bounds tightened to 8–12% and made admin-configurable within bounds	2026-09-12	Approved
CH-003	NFR-SC-02	Multitenancy (institution id) added to core tables for future campus expansion	2026-09-15	Approved

Any post-baseline change follows this log: proposal → impact analysis on RTM → supervisor approval → version increment (Appendix J).

Chapter 17 — Appendices

Appendix A — Glossary

Table A.1 — Glossary of domain terms

Term	Meaning
Campus Info Hub	The pillar that consolidates official notices, events, achievements, calendar, emergency, traffic, and weather information.
Career Path Finder	The pillar that matches a student's interests and job preference to a career track and roadmap.
Mentor-Powered Help	The pillar that lets students book verified seniors/peers for paid academic help.
Seeker / Mentor	The two roles of a paid session: the student who requests and pays; the student who delivers help.
Escrow	Funds captured at booking and released only on confirmation or admin ruling.
Commission	Platform fee (8-12%) deducted at release.
Featured placement	Paid extra visibility option for mentors in search results (future revenue line).
Roadmap	Ordered set of recommended courses, certifications, clubs, and mentor sessions toward a chosen career path.

Appendix B — Acronyms

Table B.1 — Acronyms

Acronym	Expansion	Acronym	Expansion
SRS	Software Requirements Specification	API	Application Programming Interface
SDLC	Software Development Life Cycle	REST	Representational State Transfer
SSR	Server-Side Rendering	IPN	Instant Payment Notification
RBAC	Role-Based Access Control	OTP	One-Time Password
MFS	Mobile Financial Service	UAT	User Acceptance Testing
ORM / ODM	Object-Relational / Object-Document Mapper	CI/CD	Continuous Integration / Delivery
WCAG	Web Content Accessibility Guidelines	ASVS	Application Security Verification Standard
PII	Personally Identifiable Information	RTM	Requirements Traceability Matrix
WBS	Work Breakdown Structure	MoSCoW	Must / Should / Could / Won't prioritization
PWA	Progressive Web App	MVP	Minimum Viable Product

Appendix C — Database Schema

C.1 PostgreSQL (Prisma excerpt)

```

model User {
  id          String    @id @default(uuid())
  institutionId String    @default("uiu")
  email       String    @unique
  passwordHash String
  role        Role      @default(STUDENT) // STUDENT | MENTOR | FACULTY | ADMIN
  status      Status    @default(ACTIVE)  // ACTIVE | SUSPENDED | CLOSED
  emailVerifiedAt DateTime?
  createdAt   DateTime @default(now())
  studentProfile StudentProfile?
  mentorProfile MentorProfile?
}

model MentorSession {
  id          String    @id @default(uuid())
  seekerId    String
  mentorId    String
  courseCode  String    // e.g. "CSE-1109"
  topic       String
  scheduledAt DateTime
  durationMin Int
  status      SessionStatus @default(REQUESTED)
  // REQUESTED | ACCEPTED | ESCROW_FUNDED | IN_PROGRESS
  // COMPLETED | CONFIRMED | DISPUTED | REFUNDED | DECLINED
  escrow      EscrowTransaction?
  rating      Rating?
  dispute     Dispute?
}

model EscrowTransaction {
  id          String    @id @default(uuid())
  sessionId   String    @unique
  amount      Int        // BDT, integer only
  commissionPct Decimal   @db.Decimal(4,2) // 8.00..12.00
  bkashPaymentId String   @unique
  status      EscrowStatus @default(INITIATED)
  // INITIATED | FUNDED | FROZEN | RELEASED | REFUNDED
  fundedAt    DateTime?
  releasedAt   DateTime?
  refundedAt   DateTime?
}

```

C.2 MongoDB Collections (sample document)

```

// notices
{
  "_id": ObjectId("..."),
  "type": "circular | result | achievement | seminar | grant | emergency",
  "priority": "normal | high | emergency",
  "title": "CSE-1109 Midterm Syllabus Update",
  "body": "...",
  "authorRef": { "userId": "uuid", "name": "..." },
  "audience": ["CSE", "EEE"],
  "publishAt": ISODate("2026-09-20T10:00:00Z"),
  "expiresAt": ISODate("2026-10-20T00:00:00Z"),
}

```

```

    "isEmergency": false,
    "editHistory": [ { "at": ISODate(...), "by": "uuid", "summary": "..." } ]
  }

  // mentor_public (denormalized search card)
  {
    "userId": "uuid", "name": "...", "department": "CSE",
    "courseCodes": ["CSE-1109", "CSE-2201"],
    "rate": 300, "rating": 4.7, "completedCount": 12,
    "availability": { "weekdayEve": true, "weekend": true }
  }

```

Appendix D — UML Diagrams

Figure D.1 shows the end-to-end booking and escrow sequence. Figure D.2 is the session / escrow state machine enforced server-side. Figure D.3 documents the deployment topology.

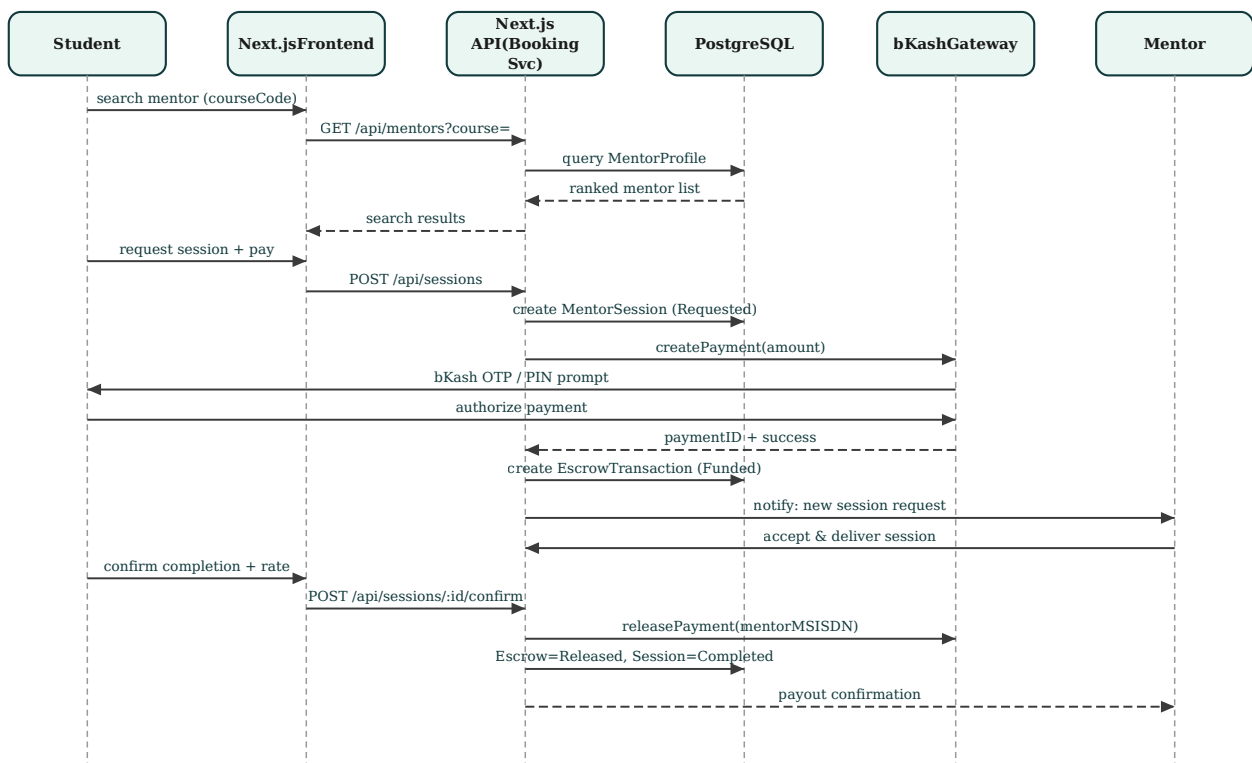


Figure D.1 — Sequence diagram: mentor booking and bKash escrow flow

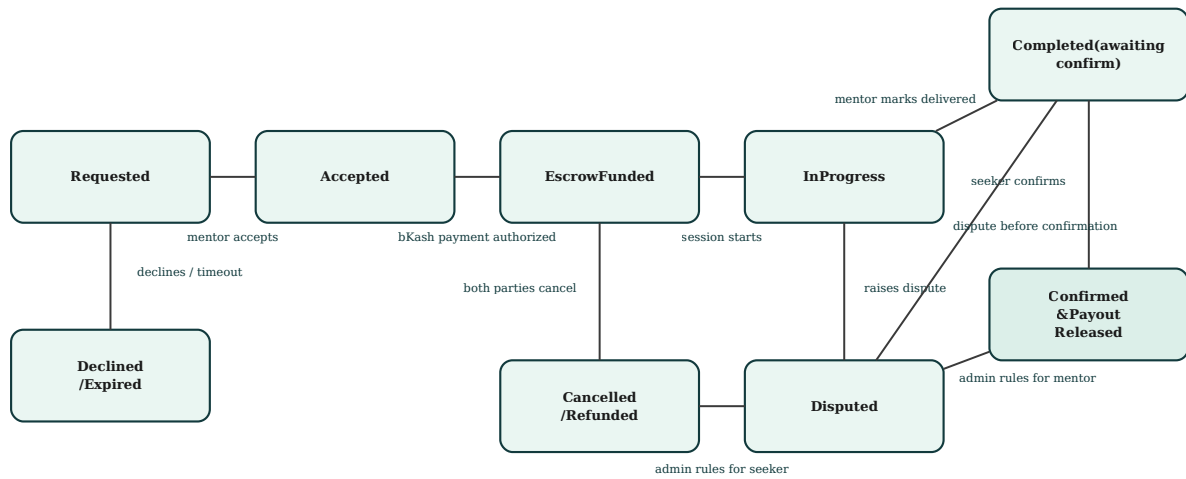


Figure D.2 — State diagram: mentor session and escrow lifecycle

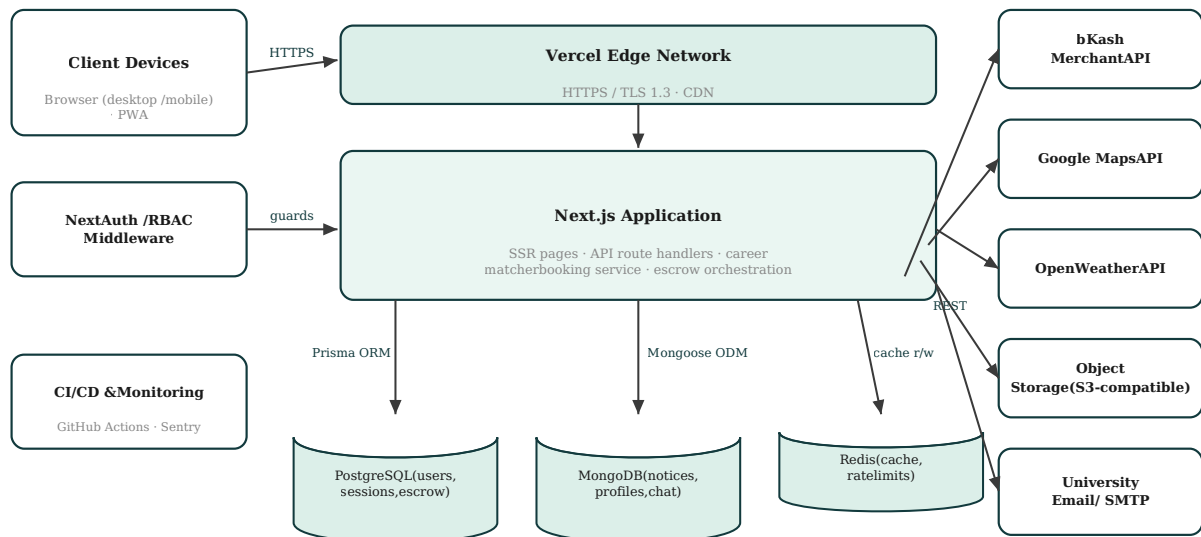


Figure D.3 — Deployment diagram of the Campus Guide platform

Appendix E — UI / Wireframes

Wireframe inventory (low-fidelity layouts delivered with the design package; screens are responsive from 360 px up):

Table E.1 — Wireframe inventory

Screen	Key Elements
Home feed	Top app bar with search; role-aware nav; emergency banner; notice cards with type/priority chips; calendar strip; traffic & weather widget row
Notice detail	Title, body, author & timestamps, audience, attachments, reminder button
Career assessment wizard	Step indicator; interest chips (multi-select); job-preference cards; progress-safe navigation

Career match results	Ranked career cards with match score & rationale; "select path" CTA
Roadmap	Vertical timeline of steps with status toggles and mentor-session suggestions
Mentor search	Course-code search box; filter rail (rating, fee, availability); mentor cards (rating, completed count, rate, availability)
Booking & payment	Slot picker; fee summary (rate × duration); escrow explainer ("held until you confirm"); bKash button
Session history	Status chips per session; confirm & rate dialog; dispute link (pre-confirmation only)
Mentor dashboard	Request queue, availability editor, earnings chart, verification status
Admin console	Verification queue; dispute panel with timeline/evidence viewer; notice publisher with emergency toggle; commission settings; analytics

Appendix F — API Documentation (Summary)

Table F.1 — Core API routes (Next.js route handlers)

Method	Route	Purpose / Notes
POST	/api/auth/register	Register with @uiu.ac.bd email (FR-UM-01); sends verification mail
GET	/api/auth/verify?token=	Activate account (FR-UM-02)
POST	/api/auth/login	Session login with rate limiting (FR-AU-01..03)
POST	/api/auth/reset	Password reset request/confirm (FR-AU-04)
GET/PUT	/api/profile	Read/update own profile (FR-PM-01..04)
GET	/api/notices?type&category&q	Feed with filters & search (FR-CI-03, FR-SF-01)
POST	/api/notices	Publish (faculty/admin; emergency flag gated by role)
POST	/api/career/assessments	Submit assessment answers (FR-CP-01)
GET	/api/career/matches?assessment=	Ranked matches with scores (FR-CP-02)
POST	/api/career/paths/:id/select	Save path & generate roadmap (FR-CP-03/04)
GET	/api/mentors?course=CSE-1109	Course-code mentor search, ranked (FR-MH-02)
POST	/api/sessions	Create session request (FR-MH-03); returns order summary
POST	/api/sessions/:id/accept /decline	Mentor decision; decline triggers refund (FR-MH-05)
POST	/api/sessions/:id/complete	Mentor marks delivered (FR-MH-06)
POST	/api/sessions/:id/confirm	Seeker confirms + rating (FR-MH-07)
POST	/api/sessions/:id/dispute	Raise dispute; freezes escrow (FR-MH-08)

POST	/api/payments/bkash/create	Create bKash payment for a session (FR-MH-04)
POST	/api/payments/bkash/ipn	Idempotent callback verification (FR-EH-03)
POST	/api/payments/release	Internal: release minus commission (FR-MH-07)
GET	/api/admin/disputes	Dispute queue (FR-AD-02)
POST	/api/admin/disputes/:id/resolve	Ruling: refund partial release; audit-logged (FR-MH-09)
PUT	/api/admin/settings/commission	Set commission within 8-12% bounds (FR-AD-03)
GET	/api/admin/analytics?from&to	Usage & revenue dashboards (FR-RP-01)

All routes: JSON bodies, session-cookie auth, RBAC middleware, consistent error envelope (FR-EH-01), schema validation at the boundary.

Appendix G — Test Cases (Expanded Payment & Dispute Suite)

Table G.1 — Expanded escrow & dispute test cases

ID	Scenario	Precondition	Expected Result
TC-16	Escrow amount matches rate × duration	Mentor rate 300/hr; 90-min session	Amount = 600 BDT; summary shown before payment
TC-17	Session without payment cannot be accepted into funded flow	Request created, bKash never completed	Session remains Requested; acceptance path blocked; timeout refund = no-op
TC-18	Seeker disputes before delivery	Funds frozen	Admin sees full timeline; ruling executes; parties notified; audit row written
TC-19	Dispute after confirmation	Session Confirmed, funds released	Dispute rejected with policy message
TC-20	Commission deduction on release	Amount 600, commission 10%	Mentor payout 540; 60 recorded as platform commission
TC-21	Refund on mutual cancellation	Funded, pre-delivery, both cancel	Full refund; state Refunded; notifications sent
TC-22	Auto-confirm after 48 h silence	Delivered, seeker inactive	Session auto-Confirms; payout released; neutral rating recorded
TC-23	Release API called twice (idempotency)	Session already Confirmed	Second call is a no-op; single payout recorded

Appendix H — Interview / Meeting Notes

Table H.1 — Consultation summary

Session	Participants	Key Findings (shaping requirements)
Student focus group (n=12)	2nd–4th year students	Notices missed on Facebook; want one verified feed and exam-week alerts (→ FR-CI-01..04); career choices made "by guesswork" (→ FR-CP-01..06); distrust of cash tutoring deals (→ escrow, FR-MH-04..09).
Mentor interviews (n=5)	Senior students	Willing to teach for a fair per-hour fee; fear non-payment and no-shows (→ FR-MH-05, FR-MH-10 auto-confirm); want visible track record (→ FR-PM-05).
Faculty consultation	Two department staff	Will publish if it is easier than managing groups; want reach metrics and no moderation burden (→ FR-CI-01, FR-RP-02).
Admin walkthrough	Prospective moderators	Dispute panel must show timeline + evidence in one view; every money action must be auditable (→ FR-AD-02, FR-AD-06).
Supervisor reviews	CSE faculty supervisor	Endorsed three-pillar scope; mandated escrow-first payment design and MoSCoW discipline (→ Chapter 7 priorities, Section 16.5).

Appendix I — Gantt Chart

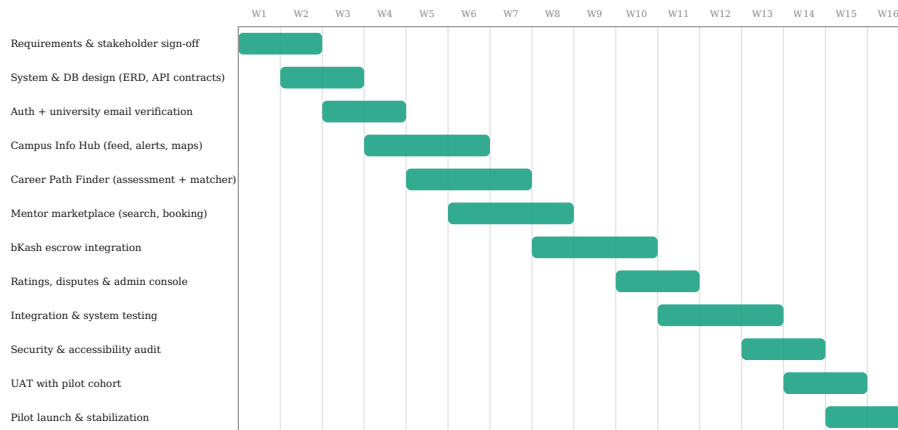


Figure I.1 — Project Gantt chart (16 weeks, 8 sprints)

Appendix J — Revision History

Table J.1 — Revision history

Version	Date	Author	Description of Change
0.1	2026-08-30	Kanij Fatema Bipa	Initial draft from the Campus Guide proposal (August 2026); chapter skeleton created.
0.5	2026-09-10	Nur Hossain Nazim	Functional requirements, use cases, and data design drafted; CH-001 incorporated.
0.8	2026-09-15	Redwan	NFRs, risk analysis, project plan, and test strategy added; CH-003 incorporated.
1.0	2026-09-22	Arafat Abir	Baseline: traceability completed, appendices A-J finalized, document approved for implementation.